

The role of phases and specificity in definite islands

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Abstract

Cross-linguistically, sentences with *wh*-dependencies across the boundary of a definite depiction nominal object are degraded. We consider two accounts – DP phasehood and the Specificity Condition – which make different predictions about definite islands in *wh*-movement and *wh-in-situ* languages, especially around whether the choice of main verb can neutralize the definite island effect. We tested these predictions with parallel experiments for English (*wh*-movement) and Mandarin Chinese (*wh-in-situ*). Confirming existing reports, we observed that both languages show a definite island effect. Additionally, in English, but not Chinese, there is a neutralization effect related to the choice of main verb. However, the effect is only partial, contrary to previous claims. A combination of the PIC and the Specificity Condition accounts for both languages, even though it would be more parsimonious to have only one constraint. To support this conclusion, we ran a third experiment on Chinese *wh*-indefinites to provide independent evidence for the Specificity Condition.

Keywords: definiteness, Specificity Condition, Phase Impenetrability Condition, *wh*-movement, *wh-in-situ*, island effect

1 Introduction

Theories of locality have recognized that *wh*-dependencies are sensitive to a variety of constraints. One such constraint that is attested across languages involves definiteness. For instance, in English, French, German, subextracting an argument out of a definite object (picture nouns) is degraded (1) (Chomsky 1973; Fiengo and Higginbotham 1981; Bowers 1987; Diesing 1992; Davies and Dubinsky 2003; Simonenko 2016; Huang 2022; Shen and Lim 2022, among many others). Similar facts have been reported in *wh-in-situ* languages as well, even though there is no subextraction involved and *wh-in-situ* is typically less sensitive to

island constraints: Mandarin Chinese, for instance, allows an *in-situ wh*-argument to appear inside a complex NP subject, if it is not marked with a demonstrative (2) or headed by a proper name (C.-T. J. Huang 1982b).

- (1) The definite island effect
- a. What did you listen to a song about?
b. *What did you listen to that song about?
- (2) a. [Ài kàn shénme de xiǎohái] méi chūxī?
love see what DE child no future
'Children who love to watch *what* have no future?'¹
b. *[Ài kàn shénme de nà-gè xiǎohái] méi chūxī?
love see what DE that-CL child no future
Intended: 'That child who loves to watch *what* has no future?' (Adapted from C.-T. J. Huang 1982b:405 exx. 140-141)

Several accounts have been proposed to explain this effect, which we will call the definite island effect.² One account claims that definite and indefinite nominals are structurally different; e.g., definite nominals are locality domains (e.g., bounding nodes or phases) but indefinite nominals are not. *Wh*-dependencies involving definite nominals are therefore subject to some principle like the Phase Impenetrability Condition (PIC; Chomsky 2000, 2001). A second account attributes this effect to an independent Specificity Condition, which blocks *wh*-dependencies (among others) from crossing the boundary of a nominal with a specific interpretation (Fiengo and Higginbotham 1981; C.-T. J. Huang 1982b, cf. Simonenko 2016). Finally, a third account appeals to information structure constraints: definite nominals render a *wh*-phrase inside it to be not dominant, thus making it incompatible with subextraction (Erteschik-Shir 1981, see also Goldberg 2006; Abeillé et al. 2020).

Logically, these accounts might all be valid, in which case the definite island effect is a composite of various constraints. However, parsimony considerations – whether by Occam's Razor or in the spirit of the Minimalist Program (Chomsky 1995) – require us to seriously consider the possibility that only one of these accounts is actually needed to explain the definite island effect. This has in fact been suggested in the literature, whether implicitly or explicitly. N. Huang (2022), for instance, analyzes a number of exceptions to the definite island effect (first discussed in detail by Davies and Dubinsky 2003) in English by treating definite nominals as DP phases. In contrast, Matushansky (2005) explicitly argues that the definite island effect cannot be taken as evidence for the phasehood of DPs, since the effect can already be accounted for by the Specificity Condition.

This paper takes a cross-linguistic and experimental approach toward addressing this question about parsimony. We focus only on the locality domain account (in particular, an account assuming the phasehood of DPs and the PIC) and the Specificity Condition, where there are relatively clear predictions of the definite island effect in both subextraction and *wh-in-situ* (we return to the information structure account in the General Discussion). We

¹Abbreviations: CL: classifier; DE: particle marking pre-nominal modifiers; PFV: perfective.

²To be sure, this effect is not necessarily found with all definite NPs. Exceptions have been reported for NPs marked by "weak" definite articles with a unique reading, bound possessors, and demonstrative articles (Erteschik-Shir 1981; Fiengo 1987; Davies and Dubinsky 2003; Simonenko 2016; Huang 2022; Lim 2022, etc.); in fact, this article will take a closer look at the case of demonstratives.

observe that both accounts actually make divergent predictions about subextraction and *wh-in-situ*, specifically, regarding whether the choice of main verb can neutralize the definite island effect. Davies and Dubinsky (2003) observe that in English at least, when the main verb denotes an action of creation, there seems to be no contrast between (3a) and (3b) (cf. similar remarks about verb choice in Erteschik-Shir 1981).

- (3) The verb of creation effect (VOC) effect
 - a. What did you compose a song about?
 - b. What did you compose that song about?

We evaluate these predictions about the VOC effect with two parallel acceptability judgment experiments on demonstrative-marked definite nominals, one looking at subextraction of *wh*-arguments in English and the other at *in-situ wh*-arguments in Mandarin Chinese. To preview our results, we find a definite island effect in both languages. However, the neutralization effect of VOCs in English is only partial, contra Davies and Dubinsky 2003 and Huang 2022. Based on these results, we argue that a theory of the definite island effect with cross-linguistic coverage requires reference to DP phasehood and the PIC (specifically building on Davies and Dubinsky 2003) as well as the Specificity Condition. In other words, both are necessary in a general theory of locality, even though a more parsimonious analysis is logically possible. We further argue that our findings of cross-linguistic variation challenge an information structure account of the definite island and VOC effects.

Finally, our account of Chinese *wh-in-situ* assumes that the interrogative reading of *wh*-arguments in general is licensed by unselective binding by a question operator, which is said to be sensitive to the Specificity Condition (Li 1992). We present a third experiment to confirm this assumption, showing that the licensing of *wh*-indefinites in a DP, where a *wh*-argument is interpreted as an indefinite “something,” is also sensitive to the specificity of that DP. This result affirms Li’s informal observations and supports her analysis of *wh*-indefinites as being bound by some operator, which in turn validates our adoption of her unselective binding account.

This paper is organized as follows. Section 2 reviews the two accounts of interest – the DP phasehood account (in particular, adapting ideas by Davies and Dubinsky 2003) and the Specificity Condition (Fiengo and Higginbotham 1981; C.-T. J. Huang 1982b) – and discusses their predictions. Section 3 presents the two experiments and results on *wh-in-situ* and overt subextraction. Section 4 presents our proposal and reports a third experiment showing that the binding of an *wh*-indefinite in Chinese is sensitive to specificity. Section 5 then discusses implications of the experimental results for information structure based theories about locality and LF movement analysis of *wh-in-situ*. Section 6 concludes.

2 Two analyses of the definite island effect

2.1 DP phasehood + Incorporation

There is a long tradition of analyzing nominals – especially definite nominals – as locality domains that block *wh*-extraction; this has been variously implemented by positing that NPs/DPs are bounding or cyclic nodes, blocking categories, or phases (Chomsky 1973, 1977, 1986a; Bach and Horn 1976; Bowers 1987; Davies and Dubinsky 2003; Huang 2022).

For concreteness, we will implement this intuition with a phase-based analysis, assuming that definite nominals are DP phases. We also adopt the following version of the Phase Impenetrability Condition (PIC) (4).

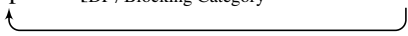
- (4) Phase Impenetrability Condition (Chomsky 2000)
 In phase α with head H, the domain of H is not accessible to operations outside of α ; only H and its edge are accessible to such operations.

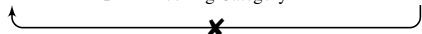
In an unacceptable sentence like **What did you listen to that song about?* the *wh*-phrase should move from the definite DP to Spec,CP, as schematically represented in (5). According to the PIC, any movement at the CP level (or vP level, assuming vP is a phase) cannot target elements inside D's domain, which includes the *wh*-phrase; it can only target elements up to D and its edge. To satisfy the PIC, then, the *wh*-phrase must first move to Spec,DP (the edge) before moving onward to a higher position.

- (5) [CP ... [DP *that song about what*]]

However, it is common to assume that Spec,DP is not available as a landing site. The absence of a landing site might be due to lexical idiosyncrasies in features: D lacks features that trigger *wh*-movement to its specifier (McCloskey 2002; Huang 2022). Alternatively, one could assume that Spec,DP is already occupied by some other constituent. For instance, a recent proposal by Jenks and Konate (2022) suggests that Spec,DP contains a covert IdXP with information about the referent of the DP. Regardless of the analysis adopted, the outcome is the same: the *wh*-phrase is denied of a landing site on the edge of the DP, causing subextraction to violate the PIC.

To explain the VOC effect, this analysis can be adapted in a few ways. Most notably, in discussing their observation that VOCs can neutralize various definite islands (including demonstrative-marked DPs) in English, Davies and Dubinsky (2003:28–29) put forward an LF noun incorporation account. They propose that the head noun of the object can incorporate onto the verb in English if the following three conditions are met: (i) the noun is a result nominal; (ii) the object is the complement of a causative verb semantically related to the denoted result (e.g. *write-book*); and (iii) the verb's subject controls (binds) the agentive subject of the object (e.g. the book's author); citing Chomsky (1986b), they further argue that demonstrative-marked DPs contain an agentive PRO which represents the creator of the result nominal and is controlled by the verb's subject (ibid.:25–27). In their account, the incorporation of the head noun onto the main verb neutralizes the locality domain (in their terms, blocking category) status of the definite DP. As a result, the DP is transparent to subextraction after incorporation is applied, as shown in (6). This noun incorporation operation, however, is not available for non-VOCs, thus the definite DP object remains a blocking category, as shown in (7).

- (6) N-incorporation with VOC: Mary composed [DP≠Blocking Category that PRO song about Sarah].


- (7) No N-incorporation with non-VOC: Mary listened to [DP=Blocking Category that PRO song about Sarah].


Strictly speaking, though, a noun directly incorporating onto the verb would violate the PIC, as the noun lies within D's domain. One solution that is compatible with the PIC while remaining largely faithful to Davies and Dubinsky's proposal is that it is the *determiner* of the definite DP that can undergo (covert) head movement to a VOC. This analysis can be seen a variant of Bošković's (2015) phase collapse account, proposed independently to explain why Galician lacks a definite island effect when the definite determiner cliticizes onto the verb shown in (8) (Uriagereka 1988). Bošković argues that in Galician, D overtly moves to V, which further moves to v, a phase head. While DP itself might lack a landing site for *wh*-movement, this sequence of head movement operations collapses the DP phase with the vP phase and makes available an alternative landing site in Spec,vP, thus ensuring that *wh*-extraction satisfies the PIC. What we suggest here is that a similar head movement operation might be available in English, except that it applies covertly under the conditions that Davies and Dubinsky (2003) proposed.

- (8) a. *e [de quén]_i viche [DP=Phase o [NP retrato *t_i*]]?
 and of who saw(you) the portrait
 b. e [de quén]_j viche-loi_i [DP≠Phase [D' *t_i* [NP retrato *t_j*]]]?
 and of whom saw(you)-the portrait
 'so, who have you seen the portrait of?' (Uriagereka 1988)

The reason for N/D-incorporation to be restricted to VOCs in English is not fully clear. Lim (2022) connects it to the fact that a VOC and its DP object are construed as describing a single event of creation (cf. the Single Event Condition for *wh*-movement proposed in Truswell 2007). For example, in *write that book*, the book is construed as the result of the writing event, while in *read that book*, the book is not construed as the result of the reading event. As support, Lim (2022) conducted experiments showing that the greater the extent a verb has creation semantics, the easier it is to subextract an argument out of the definite object DP. That being said, the Galician data in (8) show that N/D incorporation is not always restricted to VOCs. A thorough understanding of the conditions for N/D incorporation requires further research, but given the proposals cited above, we will assume that a DP phasehood account will require some incorporation mechanism, if it is to capture the above contrasts.³

For thoroughness, it should also be noted that subextraction in general is sensitive to a wider range of verbs. For example, Bach and Horn (1976) point out that subextraction from *indefinite* objects becomes unacceptable with certain verbs, examples of which are shown in (9) (see also Erteschik-Shir and Lappin 1979:74 exx. 96–97 and Kuno 1987:19ff).

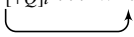
- (9) Bach and Horn 1976 exx. 34–35
 a. Who did they see/*destroy pictures of?
 b. *Who did you write/*see a book about?
 c. *Who did John burn a large green book about?

The VOC effect that this paper focuses on is the effect of VOCs on sub-extraction from definite object, on top of this general effect. Our experiment design controls for this general

³We also note that alternatives to a N/D-incorporation account are possible. For example, Huang (2022) discusses an analysis of DP islands and their neutralization that relies in part on the verb's subject transmitting its features onto the PRO inside the DP; one could assume that VOCs allow feature transmission and non-VOCs do not.

effect of verbs in (9) by including sentences with sub-extraction from both indefinite and definite objects. This allows us to highlight the effects of VOCs on sub-extraction out of definite objects, while controlling for the effect of individual verbs. We discuss this in more detail in Section 3.

To summarize, the English definite island / VOC facts can be accommodated by combining the PIC with a N/D-incorporation operation, adapting Davies and Dubinsky 2003.⁴ In a *wh-in-situ* language like Mandarin Chinese, however, this approach does not make clear predictions about the definite island effect and consequently the VOC effect. This outcome follows from the assumption that *in-situ wh*-arguments get an interrogative reading through unselective binding (Li 1992; Aoun and Li 1993; Tsai 1994, among many others). In this analysis of *wh-in-situ*, commonly adopted in the East Asian linguistics literature, *wh*-arguments are bound *in-situ* by a covert question operator in CP or some similar position in the left periphery of the clause, as schematically represented in (10).

- (10) [CP OP_[+Q]i ... what_i ...] *wh-in-situ*
- 

Since these *wh*-phrases do not undergo any movement, they are not subject to subjacency or what can be seen as its Minimalist counterpart, the PIC.⁵

The PIC approach described in this section, then, does not predict a definite island effect for *in-situ wh*-questions, contrary to judgments reported by C.-T. J. Huang (1982b). In other words, (11a) and (11b) should be equally acceptable. Note that this prediction holds even if one assumes an NP (instead of DP) structure in Chinese.

- (11) a. Zhāngsān tīng-le yì-shǒu guānyú shénme de gē?
 Zhangsan hear-PFV one-CL about what DE song non-VOC+indefinite
 lit. ‘Zhangsan heard to a song about what?’
 b. Zhāngsān tīng-le nà-shǒu guānyú shénme de gē?
 Zhangsan hear-PFV that-CL about what DE song non-VOC+definite
 lit. ‘Zhangsan listened to that song about what?’

And since the VOC effect is the neutralization of the definite island effect, such an approach will not predict a VOC effect in a *wh-in-situ* language. All else being equal, (12a) and (12b) should also be equally acceptable.

⁴A reviewer asks whether this outcome still holds if one assumes an NP structure for English where N, instead of D, is the head of the nominal domain (see Abney 1987; Bruening 2020; Chomsky 2020; Pullum and Miller 2022). If the highest projection of the nominal domain is an NP and it is a phase, the definite island effect is predicted under the PIC analysis as long as we assume that the Spec,NP of a definite NP is occupied, and thus cannot serve as a landing site, while the Spec,NP position of indefinite NPs is not occupied. This could follow if, similar to what we outlined above, there is a covert IdXP occupying the specifier position of a definite NP, adapting the proposal of Jenks and Konate (2022). With these assumptions in place, we can analyze the VOC effect as LF incorporation of N to the VOC and v. In other words, the predictions of this analysis remain the same if an NP structure is assumed.

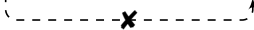
⁵As observed by C.-T. J. Huang (1982b) and many others, *in-situ wh*-adjuncts (notably those denoting reason, e.g., *wèishénme* ‘why’), unlike *wh*-arguments, seem sensitive to island constraints. Following Aoun and Li 1993; Tsai 1994, we assume that binding is not available for *wh*-adjuncts; instead, covert movement to the CP domain is involved. Although it would be theoretically interesting to test *in-situ wh*-adjuncts for definite island effects, there are several hurdles. First, reason-denoting adjuncts like *wèishénme* ‘why’ are not compatible as arguments or adjuncts of simple DPs. They must appear in a clausal structure, e.g., clausal complements or relative clauses within a DP, thus implicating a complex NP structure. However, the locality profile of complex NPs and *in-situ wèishénme* is currently under debate independent of definiteness (Lu et al. 2020; Tian et al. 2022). Second, *wèishénme* in Chinese also has a purpose reading that is claimed to be insensitive to islands, and it would be necessary but complicated to control for this interpretation. For our purposes, we will focus on *wh*-arguments for both English and Chinese.

- (12) a. Zhāngsān tīng-le nà-shǒu guānyú shénme de gē?
 Zhangsan hear-PFV that-CL about what DE song non-VOC+definite
 lit. ‘Zhangsan heard that song about what?’
 b. Zhāngsān xiě-le nà-shǒu guānyú shénme de gē?
 Zhangsan write-PFV that-CL about what DE song VOC+definite
 lit. ‘Zhangsan wrote that song about what?’

For transparency, we note that in addition to unselective binding, an alternative analysis of *wh-in-situ* questions is that the *wh*-phrase undergoes covert movement in LF (C.-T. J. Huang 1982b; Lu et al. 2020). If covert movement is indeed subject to the PIC, as Lu et al. (2020) suggest (contra standard generalizations), this analysis could predict the existence of the definite island and VOC effects in Chinese. We will discuss this approach in Section 5.2.⁶

2.2 A semantics-based approach: the Specificity Condition

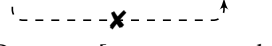
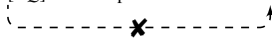
Fiengo and Higginbotham (1981) provide a semantics-based approach to the definite island effect: the Specificity Condition. This condition essentially states that a variable inside a specific DP cannot be bound by an operator outside the DP, as schematized in (13).⁷ Although Fiengo and Higginbotham do not offer a definition of what specificity is and how that relates to definiteness, Fiengo (1987) explicitly ties specificity in the context of the Specificity Condition back to the familiarity theory of definiteness. Citing Heim 1982:298, Fiengo suggests that a specific nominal is one that is already “familiar” in the discourse context while a non-specific nominal is “novel.” Crucially, the notions of familiarity and novelty are the ones traditionally associated with definiteness, such that a definite DP is typically one whose reference is “already familiar at the current stage of the conversation,” while an indefinite DP is one whose referent is unfamiliar or novel.

- (13) $OP_i \dots [_{\text{SpecificDP}} \dots \text{variable}_i \dots]$


The Specificity Condition can be seen as applying to both English and Chinese *wh*-questions, which are characterized by binding relations. In English overt *wh*-movement questions, the *wh*-element moves to Spec,CP, leaving behind a bound variable (its trace/copy), as represented in (14). In Chinese *wh-in-situ* questions, as mentioned in the previous section, a covert question operator at Spec,CP binds the *in-situ wh*-element (15). It follows straightforwardly, then, that subextraction from a definite DP in English and an *in-situ wh*-phrase inside a definite DP in Chinese both violate the Specificity Condition.

⁶For completeness, we point out that there is an alternative structural account involving the Freezing Condition (Wexler and Culicover 1980; Corver 2017, etc.). In this analysis, a definite DP always undergoes QR to a higher position, which results in the DP becoming frozen. If QR applies before *wh*-movement, then subextraction would be blocked (Mahajan 1992; Sichel 2018, also Diesing 1992). As far as we can see, this account produces very similar predictions as the DP phasehood account discussed in this section: a definite DP should be incompatible with subextraction but pose no issue for *wh-in-situ*, since the Freezing Condition affects movement but not binding. However, existing accounts are not entirely clear about what effect VOCs have, if any. Diesing (1992), in a discussion of QR, suggests that VOCs might permit only “existential closure” readings on indefinite nominal objects, with the consequence that these objects must remain *in-situ* (and can be extracted from). It is unclear whether this restriction extends to definite objects more generally and how that affects freezing. We do not want to speculate on this issue and so will leave aside the triviality approach for the rest of the paper.

⁷Fiengo and Higginbotham’s original proposal is much stronger, ruling out the binding of reflexives inside specific DPs as well. This version is too strong. It is straightforward to come up with acceptable counterexamples of reflexives inside definite DPs in English: e.g., *I_i deleted [DP that photo of myself_i].*

- (14) What ... [SpecificDP ... t_{what} ...] wh-movement

- (15) OP_[+Q]i ... [SpecificDP ... what_i ...] wh-in-situ


It should be noted that this reasoning is not novel: in fact, our reasoning here follows closely that of C.-T. J. Huang (1982b), who reports a definite island effect in Mandarin Chinese *wh-in-situ* questions, as shown in (16). Arguing that Chinese *wh*-arguments move covertly but do not violate subadjacency (which we can equate with the PIC), Huang appeals to the Specificity Condition in order to explain the definite island effect.

- (16) *Tā xiǎng-zhīdào shéi mǎi de nèi-běn shū zuì hǎo.
 he want-know who buy DE that-CL book most good
 Intended: ‘He wants to know the book that who bought is the best.’ (C.-T. J. Huang 1982b:406 ex. 148)

As further independent support for the Specificity Condition, Li (1992) claims that *wh*-indefinite readings are degraded if the *wh*-phrase occurs inside a DP marked with a demonstrative, as shown in (17). This follows if the *wh*-indefinite reading is the result of a covert existential operator binding an *in-situ wh*-phrase; the demonstrative-marked DP, being specific, blocks the binding relation. That said, anonymous reviewers expressed concerns about the reported contrast. To address such concerns, we ran an additional experiment to verify this contrast, which we report in Section 4.3.

- (17) a. Wǒ yǐwéi tā ná-le [non-specificDP shénmè rén dè xiàngpiàn].
 I think he take-PFV what man DE picture
 ‘I thought he took away a picture of someone.’
 b. *Wǒ yǐwéi tā ná-le [SpecificDP nà-zhāng shénmè rén dè xiàngpiàn].
 I think he take-PFV that-CL what man DE picture
 Intended: ‘I thought he took away that picture of someone.’ (Li 1992:148 ex. 54, our translations)

As far as we can tell, all existing variants of the Specificity Condition are silent regarding the VOC effect. As formulated, it is the specificity of the DP that drives the condition; the type of verb that takes the DP as its object should not have any impact on the specificity of the DPs. For this reason, the Specificity Condition does not predict the definite island effect to weaken or disappear in the presence of a VOC in either English or Chinese.

At this point, we should note that the term “specificity” also surfaces in influential work by Cheng and Sybesma (1999) on Chinese nominal expressions, although we believe that there, “specificity” refers to a semantic property that is related but not identical to what Fiengo and Higginbotham have in mind. In particular, for Cheng and Sybesma 1999 (and Sybesma 1997), “specificity” and “definiteness” are distinct concepts, so one can meaningfully talk about “nonspecific indefinite” and “specific indefinite” nominals, for instance. Notably, whether an indefinite nominal is “specific” or “nonspecific” depends on the kind of predicate it appears with. Bounded predicates (e.g., VPs with a perfective reading), for instance, force a “specific”

reading onto an indefinite nominal. As far as we can tell, however, Cheng and Sybesma’s conceptualization of specificity does not critically depend on notions like familiarity or novelty, unlike Fiengo’s.

There is also empirical reason to think that these two discussions are referring to different notions of the term “specificity.” In the literature on subextraction, examples illustrating acceptable subextraction from indefinite objects often feature bounded predicates in the past tense as shown in (18). In Cheng and Sybesma’s terms, these indefinite objects are presumably also specific. But if this is the same kind of specificity that the Specificity Condition refers to, these examples should be unacceptable. Furthermore, to preview Sections 3 and Section 4.3, our experiments in English and Chinese used bounded predicates in all conditions. We were able to consistently detect an acceptability contrast between indefinite and definite objects.

- (18) a. Who did John buy a book about? (Erteschik-Shir 1981:152 ex. 17d)
b. Who did you see a picture of? (Fiengo 1987:163 ex. 2b)

We should emphasize that we are not trying to deny the validity of Cheng and Sybesma’s notions of “specificity” or “definiteness.” Rather, our goal is to point out an ambiguity in how “specificity” is used, and to clarify which sense we have in mind for the purpose of this article.

Finally, we would like to point out a conceptually similar analysis of the definite island effect that appeals to some kind of semantic or pragmatic oddness. For instance, Simonenko (2016) observes that definite DPs are referential. Questioning part of that DP would yield a semantically trivial question, thus inducing unacceptability. Chaves and Putnam (2020) propose an analysis of the same nature: a question like *Who did you take that picture of?* implies that the asker knows the reference of the picture but not the individual depicted, which is pragmatically strange. Regarding *wh*-questions, this approach makes the same set of predictions as the Specificity Condition: the definite island effect should be observed in both *wh*-fronting questions in English and *wh*-in-situ questions in Chinese. Like the Specificity Condition, it is silent about the effect of verb choice, and so makes no clear prediction about the VOC effect.

However, this analysis differs from the Specificity Condition in covering only the definite island effect in *wh*-questions, while the Specificity Condition, as formulated by Fiengo and Higginbotham (1981), covers a wider range of phenomena, including *wh*-questions, relativization, focus and topic movement, as well as unselective binding. As we show later in Section 4.3, we can observe a definite island effect for Chinese *wh*-indefinites (unselective binding). The same effect can also be found in English relativization, as reported by Ross (1967) and Simonenko (2016) (contra Grosu 1973) and experimentally verified in Shen and Lim (2022). Although these findings do not argue against the triviality analysis for *wh*-questions, they show that the Specificity Condition covers more empirical ground. We therefore leave this approach aside for the rest of the article.

2.3 Predictions

The previous section reviewed two approaches proposed for the definite island effect: a DP phasehood approach, supplemented with a N/D-incorporation analysis, and the Specificity Condition. While they converge in predicting a definite island effect for an overt *wh*-movement language like English, we also noted that the approaches diverge in several ways. Table 1 summarizes the predictions. The DP phasehood and N/D incorporation approach predicts the definite island effect (for non-VOCs) and the VOC effect in English

but not in Chinese. In contrast, the Specificity Condition predicts the definite island effect in both English and Chinese, although it does not make claims about the effect main verbs might have. All else being equal, though, it is reasonable to infer from existing discussions of this condition that the choice of verbs is unlikely to matter and so there should be no VOC effect.

Table 1: Predictions

	DP phasehood+Incorporation	Specificity Condition
Definite island effect in English	non-VOCs only	Both non-VOCs and VOCs
VOC effect in English	Yes	No (all else being equal)
Definite island effect in Chinese	No	Both non-VOCs and VOCs
VOC effect in Chinese	No	No (all else being equal)

We further pointed out that these predictions are not fully consistent with existing generalizations in the literature. For instance, the DP phasehood approach predicts that Chinese should lack the definite island effect, contrary to claims by C.-T. J. Huang (1982b), while the Specificity Condition does not make clear predictions about the VOC effect, contrary to claims by Davies and Dubinsky (2003) and Huang (2022).

However, it is worth taking a second look at these claims. C.-T. J. Huang’s (1982b) observations about Chinese *in-situ wh*-arguments being insensitive to complex NP islands have been recently questioned by Lu et al. (2020) (but see Tian et al. 2022; Jin and Yan 2024; Yue and Xiang 2025). One could in turn wonder about the status of C.-T. J. Huang’s observations about definite islands. Second, one might also have concerns over the actual strength of the VOC effect in English. For instance, Brian Agbayani (p.c.) has suggested to us that subextraction from demonstrative-marked DPs is not fully acceptable even when a VOC is present. Similarly, in an experimental study, Huang (2022, Experiment 4) reports that subextraction from demonstrative-marked DPs like (19a) was judged as relatively less acceptable than minimally-different indefinite baselines, such as (19b) (median ratings of 6 vs. 7 on a 7-point Likert scale, a difference that was statistically significant in a cumulative link mixed-effects model).

- (19) a. A few people asked which old hotel Alex wrote that article about.
b. A few people asked which old hotel Alex wrote an article about.
(from Huang 2022:449 ex. 39)

For these reasons, we took an experimental approach toward evaluating these predictions about the definite island and VOC effects. Although there has been experimental work confirming the existence of the definite island effect in English (Cowart 1997; Keller 2000; Tollan and Heller 2015; Neal and Dillon 2021; Shen and Lim 2022), no such work has been done on the definite island effect in *wh-in-situ* languages in general or in Chinese in particular. Moreover, while the VOC effect has been reported in various papers in English, it has rarely been experimentally verified (but see Lim 2022).⁸ In the following section, we present

⁸Lim (2022) reports experiments comparing subextraction out of English possessive DPs with VOCs and non-VOCs as shown in (i). Her results show that VOCs improve subextraction when compared with non-VOCs. However, there is no direct comparison between subextraction from an indefinite DP and a definite DP, as all her conditions featured possessive DPs (presumably definite).

two experiments – one each for English and Mandarin Chinese – aimed at verifying this effect and comparing this effect in both overt *wh*-movement and *wh-in-situ* languages, taking English and Chinese as representatives of these two types of languages. For scope reasons, our experiments will consider only *wh*-movement of arguments and *in-situ wh*-arguments, leaving *wh*-adjuncts for future research. This is primarily so that we can engage directly with the existing literature on the VOC effect, DP phasehood, and Specificity Condition, which are typically based on *wh*-argument examples. We discuss issues related to testing *wh*-adjuncts in Section 5.3.

3 Experiments 1 and 2

3.1 Design

We report both English and Mandarin experiments together in this section, as they were designed to be parallel to each other. Our experiments adopt a factorial definition of island effects standard in the experimental syntax literature (see, among many others, Sprouse et al. 2016; Lu et al. 2020; Tian et al. 2022; Shen and Lim 2022; Jin and Yan 2024). Traditionally, for instance, a definite island effect is informally understood as the unacceptability due to subextraction from a definite DP, reflecting a constraint violation. However, in principle, the unacceptability might be attributable not to a constraint violation, but to just processing difficulty due to two non-grammatical factors: subextraction, which produces a relatively long *wh*-dependency, and definiteness (see, e.g., Hofmeister and Sag 2010 and references therein). Both factors might independently reduce acceptability so much so as to produce a perception of unacceptability. The advantage of the factorial design is that it controls for these factors, and so allows us to estimate by how much each factor reduces acceptability. If there is indeed a constraint violation with subextraction from a definite DP, then we expect to further detect a reduction in acceptability above and beyond the reduction associated with each of these factors.

In the present context, there are three factors that could independently impact the acceptability of subextraction of *wh*-arguments/*in-situ wh*-arguments: the length of the *wh*-dependency, the definiteness of the DP object, and the choice of the verb taking that DP as an object. Correspondingly, we used a 2×2×2 factorial design, with eight conditions. As shown in (20), all eight conditions in English featured embedded *wh*-interrogatives and a DP object containing the preposition *about*, so that any variation in acceptability cannot be attributed to clause type or our choice of preposition.⁹ The *wh*-interrogatives in turn were the complements of the predicate *want(s) to know*, which takes interrogative clausal complements. Within the *wh*-interrogatives, we manipulated whether there is subextraction from a DP object (if there is, the *wh*-phrase, in this case the complement of the preposition *about*, is *what*; otherwise,

To evaluate claims about the VOC effect more thoroughly, our experiments compare subextraction out of indefinite and definite DPs with VOCs and non-VOCs.

- | | | | |
|-----|----|---------------------------------------|--------------------|
| (i) | a. | What did Sally snap her picture of ? | (VOC, ex. 71b) |
| | b. | What did Sally share her picture of ? | (non-VOC, ex. 71d) |

⁹An anonymous reviewer asked why we did not use the English preposition *of*, which also features frequently in examples of subextraction, e.g., *Who did you see a photo of?* We used *about* and not *of* because *about* has a clear Chinese equivalent in *guānyú*, which is the preposition used in our Chinese materials. In contrast, there is no obvious equivalent for *of*. In Chinese, for instance, there is no structural equivalent to *a photo of Mary*: the only natural way to describe a photo depicting Mary is to say *Mǎlì de xiàngpiàn* ‘Mary DE photo’ or ‘Mary’s photo.’

the *wh*-phrase, in this case the subject of the VOC or non-VOC verb, is *who*); whether the DP object is indefinite (marked with *a(n)* in English) or definite/specific (marked with the demonstrative *that* in English); and whether the verb was a VOC or a non-VOC.

- (20) a. non-VOC *hear (joke)*
 (i) Everyone wants to know who heard a joke about the president. (ind | short)
 (ii) Everyone wants to know what Matt heard a joke about. (ind | long)
 (iii) Everyone wants to know who heard that joke about the president. (def | short)
 (iv) Everyone wants to know what Matt heard that joke about. (def | long)
 b. VOC *tell (joke)*
 (i) Everyone wants to know who told a joke about the president. (ind | short)
 (ii) Everyone wants to know what Matt told a joke about. (ind | long)
 (iii) Everyone wants to know who told that joke about the president. (def | short)
 (iv) Everyone wants to know what Matt told that joke about. (def | long)

With this 2x2x2 design, we calculated two difference-in-difference (DD) scores, which let us quantify the reduction in acceptability (if any) above and beyond the factors manipulated (see Sprouse et al. 2016; Shen and Lim 2022 for English; Lu et al. 2020; Tian et al. 2022; Huang 2024 for Chinese). The first DD score, defined as $((20a-iii) - (20a-iv)) - ((20a-i) - (20a-ii))$, represents the difference between two difference scores. The first difference score, $((20a-i) - (20a-ii))$, serves as a baseline, quantifying the impact on acceptability due to subextraction from an indefinite DP when the verb is a non-VOC. The second difference score, $((20a-iii) - (20a-iv))$, quantifies the impact on acceptability of subextraction from a definite DP for a non-VOC verb. We then compared the two difference scores, to see how much larger the impact is for definite DPs.

The second DD score we calculated is defined as $((20b-iii) - (20b-iv)) - ((20b-i) - (20b-ii))$. The logic again is the same as before, except here it quantifies the effects on acceptability in the context of VOC verbs.

Supposing there is some constraint against subextraction from definite DPs (i.e. definite DPs are islands), then by definition, the difference score for definite DPs should be larger than that for indefinite DPs, so DD scores should be positive. However, to the extent VOCs can ameliorate the definite island effect, subextraction from definite DPs in the presence of a VOC should be more acceptable, perhaps even comparable to subextraction from indefinite DPs. If so, the DD score for VOCs should be smaller than the DD score for non-VOCs, if not close to zero. In other words, the difference between the two DD scores measures the acceptability impact of verb class (VOC vs. non-VOC) on subextraction from definite DP objects (relative to an indefinite DP object baseline).

To ensure comparability, we used the same 2x2x2 design/eight conditions for the Chinese experiment, manipulating exactly the same three factors, except that the *wh*-dependency manipulation involved whether the *in-situ wh*-phrase appears outside or inside the DP object, and the definiteness manipulation contrasted the indefinite *yī* “one” with the demonstrative *nà* “that.” The eight resulting conditions are shown in (21).

- (21) a. non-VOC *tīng* (*xiàohuà*) ‘listen to/hear (joke)’
- (i) Dàjiā xiǎngzhīdào shéi tīng-le yí-gè guānyú zǒngtǒng de xiàohuà.
 everyone wonder who hear-PFV one-CL about president DE joke
 ‘Everyone wonders who heard a joke about the president.’ (ind | short)
- (ii) Dàjiā xiǎngzhīdào Shěn-lǎoshī tīng-le yí-gè guānyú shénme de xiàohuà.
 everyone wonder Shen-teacher hear-PFV one-CL about what DE joke
 ‘Everyone wonders what Teacher Shen heard a joke about.’ (ind | long)
- (iii) Dàjiā xiǎngzhīdào shéi tīng-le nà-gè guānyú zǒngtǒng de xiàohuà.
 everyone wonder who hear-PFV that-CL about president DE joke
 ‘Everyone wonders who heard that joke about the president.’ (def | short)
- (iv) Dàjiā xiǎngzhīdào Shěn-lǎoshī tīng-le nà-gè guānyú shénme de xiàohuà.
 everyone wonder Shen-teacher hear-PFV that-CL about what DE joke
 ‘Everyone wonders what Teacher Shen heard that joke about.’ (def | long)
- b. VOC *jiǎng* (*xiàohuà*) ‘say (joke)’
- (i) Dàjiā xiǎngzhīdào shéi jiǎng-le yí-gè guānyú zǒngtǒng de xiàohuà.
 everyone wonder who say-PFV one-CL about president DE joke
 ‘Everyone wonders who told a joke about the president.’ (ind | short)
- (ii) Dàjiā xiǎngzhīdào Shěn-lǎoshī jiǎng-le yí-gè guānyú shénme de xiàohuà.
 everyone wonder Shen-teacher say-PFV one-CL about what DE joke
 ‘Everyone wonders what Teacher Shen told a joke about.’ (ind | long)
- (iii) Dàjiā xiǎngzhīdào shéi jiǎng-le nà-gè guānyú zǒngtǒng de xiàohuà.
 everyone wonder who say-PFV that-CL about president DE joke
 ‘Everyone wonders who told that joke about the president.’ (def | short)
- (iv) Dàjiā xiǎngzhīdào Shěn-lǎoshī jiǎng-le nà-gè guānyú shénme de xiàohuà.
 everyone wonder Shen-teacher say-PFV that-CL about what DE joke
 ‘Everyone wonders what Teacher Shen told that joke about.’ (def | long)

We used the English distal demonstrative *that* and Chinese distal demonstrative *nà* (along with a classifier) to mark definiteness, and English *a(n)* and Chinese *yí* ‘one’ to mark indefiniteness. We chose demonstratives as our markers of definiteness for several reasons. First, Davies and Dubinsky’s account explicitly claims that subextraction from demonstrative DP objects with verbs of creation is grammatical, so using demonstratives let us directly test this claim. An alternative option would have been the definite article, like English *the*, but as the literature on definiteness has shown, subextraction facts for the definite article *the*

depend on what reading it has: subextraction is degraded under the anaphoric reading but not under the uniqueness reading (Fiengo 1987; Simonenko 2016; Shen and Lim 2022 and references therein). Second, using *the* would have presented practical challenges for creating suitable Chinese experimental materials, since Chinese lacks overt definiteness markers with *the* semantics. An anonymous reviewer asked whether our choice of demonstratives might present complications, as English *that* is ambiguous between a deitic and an anaphoric reading.¹⁰ For example, Chaves and Putnam (2020:65) suggests that the unacceptability of subextraction out of demonstrative DPs in English is due to the deictic reading of *that*. However, it is not the case that the anaphoric reading is more compatible with subextraction. According to Simonenko 2016:662, this reading also blocks subextraction. We therefore do not expect the ambiguity to pose complications.

The same reviewer also pointed out that the English and Chinese materials are not structurally identical to each other, and asked if this might introduce confounds for the cross-linguistic comparison. In particular, the Chinese DP objects we use in our experiment are of the sequence determiner–PP–DE–N e.g., “that about president *de* joke.” The particle *de* between the PP “about president” and the head noun is also used between relative clauses and head nouns in Chinese as shown in (22). The reviewer suggests that the Chinese DPs we tested might actually have a relative clause-like modifier, whereas the English DPs do not. Thus we might detect a complex NP island effect in addition to the definite island effect in Chinese.

- (22) Wǒ xǐhuān [_{RC} Zhāngsān xiě de] shū
 I like Zhangsan write DE book
 ‘I like the book that Zhangsan wrote.’

We note that the majority of existing experimental work show that Mandarin Chinese *in-situ wh*-arguments do not show complex NP island effects; see Section 5.2 for discussion. Even if the Chinese DPs are not complex, the reviewer worries that a *de*-marked DP might be hard to process independently. For instance, perhaps the determiner–modifier–DE–N order (“that about president *de* joke”) in Chinese is harder to process than the determiner–N–PP order in English (*a joke about the president*), given the greater distance between the determiner and the head noun in Chinese.

While we acknowledge the differences between the two languages, we think these differences are unlikely to compromise our conclusions. In our Chinese experiment, all conditions featured *de* and determiner–modifier–DE–N order, which would affect acceptability in every condition. For our purposes, island effects are determined not by the ratings of any individual condition, but by DD scores. By definition, calculating DD scores requires taking the difference in acceptability between conditions, in which case any effect related to *de* and word order should be canceled out. For instance, even if the determiner–modifier–DE–noun sequence is hard to process, these difficulties would be reflected in the acceptability ratings of both indefinite DPs and definite DPs. This effect would be canceled out when we calculate the acceptability differences between these two types of DPs. Consequentially, the DD scores, which are in turn based on the acceptability differences calculated, would be free of the effect of this particular processing difficulty.¹¹

¹⁰Note that the same ambiguity exists for Chinese demonstratives, such as the distal *nà* “that.”

¹¹See the appendix for a follow-up experiment addressing another concern raised by the reviewer that definiteness and the structure entailed by *de* interact to induce what looks like an island effect (a super-additive interaction effect).

3.2 Materials

In order to create the English target items, we first created 24 distinct sentence frames, combining 20 creation and non-creation verbs (10 each) with 23 nouns as the head noun of the verb's object. Each frame includes one noun and two verbs (one VOC and one nVOC), an example of which is shown in (23).

- (23) a sample sentence frame (N: *TV show*, nVOC: *watch*, VOC: *direct*)
- a. Amy watched a TV show about superheroes
 - b. Amy directed a TV show about superheroes

Our nouns are all content nouns (e.g., *joke*, *book*, *song*, etc.), which have been extensively discussed in the literature. These content nouns in turn were paired with a PP containing the preposition *about* and an DP. We then created eight versions of each sentence frame, one per condition, producing 192 sentences. These sentences were then sorted into eight different lists using a Latin Square design. In each list, each of the 24 frames appeared only once, and there were three instances of each condition.

To each list, we added 52 filler sentences. Four of these were attention checks. For two of them, participants were explicitly told to click on the left or the right end of the slider used for rating acceptability (see Section 3.4). The other two consisted of a grammatical sentence and an ungrammatical sentence with a subject-verb agreement error, whose acceptability are not in doubt. The remainder fillers all featured embedded *wh*-interrogatives, of which 15 were grammatical and acceptable and the 33 were intended to be ungrammatical and unacceptable, containing errors related to word order and subject-verb agreement, and so on). Assuming that three of the 24 target sentences are unacceptable (namely, subextraction from definite DPs with non-VOCs), this distribution of fillers would ensure that participants encounter an equal number of acceptable and unacceptable sentences. Consequently, participants would be less likely to adjust their responses so that they can distribute their responses equally across the rating scale.

It should be noted that three of our English verbs (*hear*, *read*, *write*), can appear without objects and with *about*-PPs. This fact raises the possibility that sentences with these verbs and objects are syntactically ambiguous – an *about*-PP could be attached either to the DP (24a) or the VP (24b), with potential complications for subextraction (Bach and Horn 1976, but see Erteschik-Shir 1981; Davies and Dubinsky 2003 for counterarguments). We ran a post-hoc analysis excluding data for the sentences containing these verbs; see Section 3.6 for details.

- (24) a. She [_{VP} wrote [_{DP} the book [_{PP} about dinosaurs]]].
- b. She [_{VP} wrote [_{DP} the book] [_{PP} about dinosaurs]].

We repeated the same process to create the Chinese materials, by creating 24 distinct sentence frames with 24 verbs and 24 nouns. The verbs and nouns were chosen so that the frames are semantically similar to their English counterparts.¹² The preposition we used,

¹²We note that the Chinese experiment include 1 noun and 4 verbs more than the English experiment. This is because we were able to come up with more lexical variants in Chinese for their English counterparts. The differences are listed in (i), all of which involve one variant in English corresponding to two variants in Chinese. As can be seen, the semantics of the nouns and verbs remains very similar between the two languages.

(i) a. English: *song*; Chinese: *gē* and *gēqǔ*, both 'song'

b. English: *shoot* (a video); Chinese: *pāi* and *pāishè*, both 'shoot'

guānyú, was also deliberately chosen to semantically match English *about*. Our Chinese materials also featured 52 filler sentences, including 4 attention check trials. Although the mix of acceptable and unacceptable fillers was the same, we note that filler sentences were not translated or structurally identical to their English counterparts. The test items for Experiments 1 and 2 are available at https://osf.io/9au5x/overview?view_only=656d01f7f4a7436b9276f55615a35411.

Although we ensured that our English and Chinese materials matched each other in terms of length of *wh*-dependencies, definiteness, and verb and noun choices (i.e. the factors of interest), an anonymous reviewer pointed out that our materials were not always perfectly matched in the DP complement of the preposition *about* or *guānyú* in the short condition, which corresponds to the *wh*-phrase in the long condition. There is sometimes a mismatch in animacy, in that the *wh*-phrase is invariably *what* and *shénme* “what,” but the DP complement in the short condition sometimes has human referents (e.g., . . . *who heard that joke about the president* vs. . . . *what Matt heard that joke about*).

However, we believe that these mismatches are unlikely to substantially affect our results. Any such effect should be neutralized due to the DD score analysis proposed here. For the sake of argument, suppose that animate DP complements of *about* and *guānyú* are indeed somehow more acceptable than inanimate ones, including *what* and *shénme*. In the context of a set of sentences like (20), this would mean that each of the four short conditions would be more acceptable than expected than their long condition counterparts. Since the DD score formula is aimed at quantifying the differences between long-short pairs, the fact that the mismatch appears for each long-short pair means that any acceptability effect due to the mismatch would get canceled out.

3.3 Participants

For English, 56 participants were recruited over Prolific, a crowdsourcing website. All participants were self-reported native speakers of English. Each participant received GBP 2.50 upon completing the experiment.

For Chinese, 56 participants were recruited over Prolific and from the National University of Singapore (NUS) community. Prolific participants were also self-reported native Mandarin speakers. Each Prolific participant received GBP 3 upon completing the experiment, to incentivize Mandarin speaker participation, as Mandarin speakers are relatively rare on Prolific. Each NUS participant received SGD 5 upon completing the experiment.

3.4 Procedure

The experiment was hosted on PCIBex (Zehr and Schwarz 2018), which participants accessed over their internet browsers. Participants were instructed to judge the sentence using their own intuitions as native speakers and indicate acceptability using a 100-point slider. Participants were not shown what value they selected, although they could presumably make a good inference by visual inspection. Participants also saw three practice sentences before starting the experiment proper. These practice sentences were intended to be highly acceptable, highly

-
- c. English: *make* (a short video); Chinese: *zuò* ‘make’ and *lù* ‘record’
 - d. English: *compose* (a song); Chinese *zuò* ‘compose’ and *chuàngzuò* ‘create, compose’
 - e. English: *hear* (a joke/an anecdote); Chinese: *tīng* ‘hear (a joke, song, music)’, *tīngshuō* ‘hear (an anecdote)’

unacceptable, and of marginal acceptability (in that order). Participants were explicitly told about their acceptability.

3.5 Data analysis

We standardized participant ratings using the z-score transformation in order to control for variation in how each participant used the slider scale. We also excluded 3 participants from the English experiment and 3 participants from the Chinese experiment; these participants failed at least one of the four attention checks (e.g., clicking on the right end of the scale when they should have clicked on the left end instead).

Ratings for the eight conditions were used to calculate the two difference-in-difference (DD) scores that measure the impact of verb class on subextraction from definite DPs.

For target items, we model z-scored acceptability ratings using linear mixed effects models with the R packages lme4 (Bates et al. 2015) and lmerTest (Kuznetsova et al. 2017). Unless otherwise stated, the model’s predictors were the three factors – definiteness, dependency length, and verb class (we coded indefinites as 0, definites as 1; short dependencies as 0, long dependencies as 1; non-VOCs as 0, VOC as 1). We started with maximal models with by-subject and by-item random intercepts and slopes. Random terms were removed from the model with maximal random effects until convergence.

3.6 Results

Mean acceptability ratings are displayed in Figure 1.

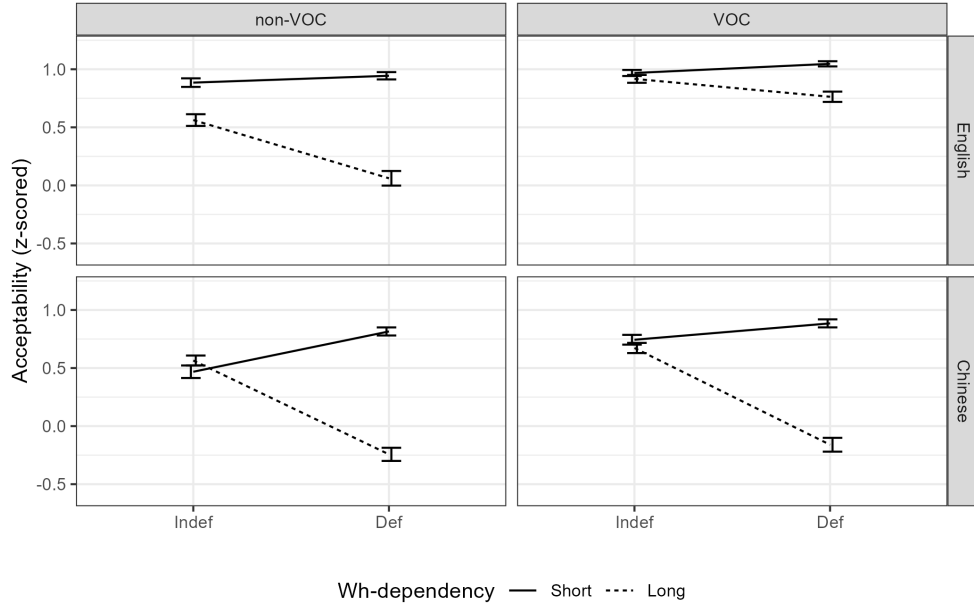
We first discuss the English results. With non-VOC main verbs, subextraction reduced acceptability for both indefinite and definite DPs, although the degree of reduction was greater for definite DPs, as reflected in the DD score of 0.56. With VOC main verbs, the reduction in acceptability was milder, as reflected in the smaller DD score of 0.23.

The difference between the two DD scores indicates that VOCs do improve subextraction from definite DPs in English. Our linear mixed effects analysis found a positive three-way interaction between VOCs, definiteness, and long *wh*-dependency ($\beta = 0.32$, $s.e. = 0.10$, $t = 3.21$, $p < .01$). We take this to be evidence of a VOC effect as reported in the literature.

That said, the fact remains that in the VOC case, subextraction from definite DPs still yielded a positive DD score (0.23). A second linear mixed effects model of the VOC acceptability data was constructed with definiteness and *wh*-dependency length as predictors (with the same coding system as the previous model), as well as by-participant random intercepts and random slopes for definiteness and length, and by-item random intercepts and random slopes for length (following the same procedure of starting with the maximal model and reducing the random effects until convergence). This second model found a significant negative two-way interaction between definiteness and long *wh*-dependencies within the VOC data set ($\beta = -0.23$, $s.e. = 0.06$, $t = -3.99$, $p < .001$). This result indicates that for demonstrative DPs, VOCs only reduce the definite island effect but do not completely neutralize it, contra Davies and Dubinsky 2003 and Huang 2022.

As for Chinese, with non-VOC main verbs, *in-situ wh*-phrases are worse in definite DPs, as reflected in the DD score of 1.15. With VOC main verbs, the DD score is 0.97. A linear mixed effects analysis of these results (following the same procedure and coding system as

Fig. 1: Acceptability of subextraction for English and *wh-in-situ* for Chinese



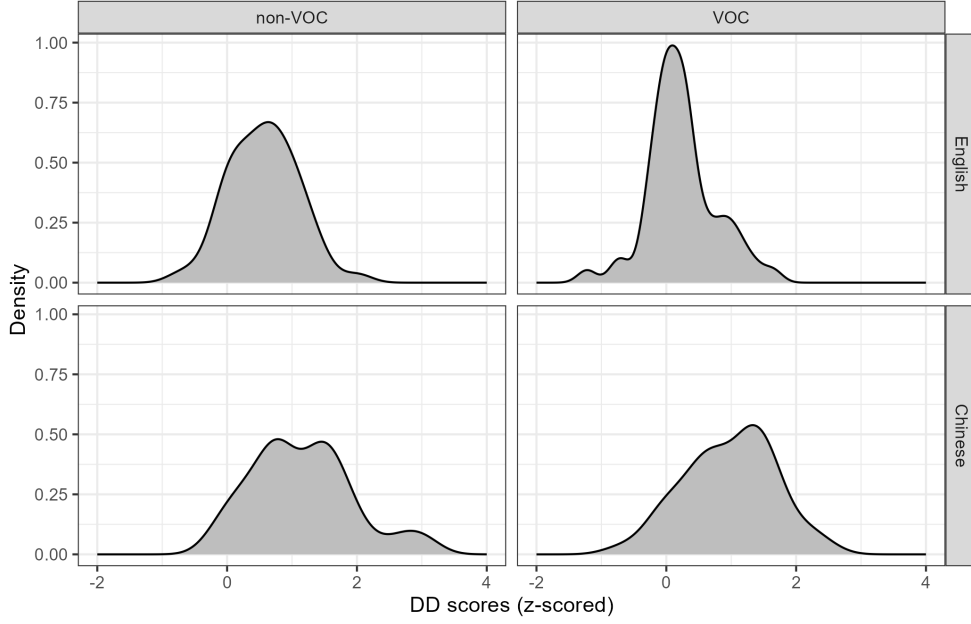
Note: For reference, in terms of mean z-scored acceptability ratings, our English filler items had a range of -1.36 to 0.86 ; unacceptable fillers had a mean rating of -0.69 and acceptable fillers had a mean of 0.28 . For Mandarin, the range was -1.26 to 1.06 , and unacceptable fillers had a mean of -0.60 and acceptable fillers a mean of 0.69 .

the models above) confirmed that the DD scores are different from 0: first, there was a negative interaction between definiteness and long *wh*-dependencies (*in-situ wh*-phrase inside a definite DP, $\beta = -1.16$, $s.e. = 0.11$, $t = -10.28$, $p < .001$). Second, there was no three-way interaction between VOCs, definiteness, and long *wh*-dependencies ($\beta = 0.19$, $s.e. = 0.11$, $t = 1.73$, $p = 0.09$), indicating that the two DD scores are not significantly different from each other. Overall, these results show that Chinese has a definiteness effect but unlike English, lacks a VOC effect.¹³

A reviewer also asked whether there were differences across participants. This is particularly relevant in the case of English VOCs, with a small but positive DD score that challenges existing claims in the literature. Perhaps the low score is due to bimodality—the existence of two distinct groups of participants, one for whom subextraction is highly acceptable, and another for whom subextraction is highly unacceptable. To address this possibility, we calculated DD scores for each participant. The distribution of DD scores by language and verb class are shown in Figure 2. For English VOCs, there is no clear evidence of bimodality; DD scores are clustered quite clearly unimodally. Similar unimodal distributions are also observed for the other language–verb class combinations.

¹³Related to the discussion of (22), we note that in the Chinese experiment the indefinite conditions were judged almost the same regardless whether the *in-situ wh*-element is inside the object DP or not. This result indicates that the DP objects containing *de*, whether modified by a relative clause or not, do not impose a noticeable processing difficulty that drives the acceptability down.

Fig. 2: Distribution of DD scores for English and *wh-in-situ* for Chinese



Finally, we ran a follow-up analysis to address a potential issue in our English materials. Recall that we used three verbs – *hear*, *read*, and *write* – where target sentences with these verbs are in principle syntactically ambiguous. Out of an abundance of caution, we re-ran the analysis on the responses to sentences that did not contain these verbs. Results were qualitatively the same as those for the larger data set. The DD score for non-VOCs is 0.53, larger than the DD score of 0.23 for VOCs. Statistical analysis again confirmed that these two DD scores are significantly different, indicating a VOC effect: a linear mixed effects model of acceptability with by-participant random intercepts and random slopes for definiteness, length, and verb, as well as by-frame random intercepts and random slopes for the same predictors, detected a significant positive three-way interaction between VOCs, definiteness, and long *wh*-dependencies (a VOC effect, $\beta = 0.30$, $s.e. = 0.11$, $t = 2.68$, $p < .01$). We also confirmed that the DD score of .23 is statistically different from 0, i.e. there is a definite island effect even with VOCs: a model fitted on the VOC acceptability data with by-participant random intercepts and random slopes for the main effects and their interaction, as well as by-item random intercepts and slopes for definiteness and *wh*-dependency, found a significant negative two-way interaction effect between definiteness and long *wh*-dependencies (a definite island effect, $\beta = -0.23$, $s.e. = 0.10$, $t = -2.38$, $p < .05$).

4 Proposal

4.1 DP phasehood + Specificity Condition

Table 2 summarizes our experiment results and compares them against the predictions from the DP phasehood approach and the Specificity Condition. As one can see, neither approach alone accounts for the full range of results. The DP phasehood approach does not predict the presence of the definite island effect in Mandarin Chinese, as the PIC only restricts movement, and not binding. Furthermore, the DP phasehood approach cannot account for the presence of a definite island effect in English even when the main verb is a VOC. In this approach, the head noun or D should have incorporated onto the VOC, neutralizing the PIC completely. On the other hand, the smaller definite island effect under VOCs in English is puzzling for the Specificity Condition approach, which is silent about the effect of the choice of verb. In other words, our experiments show that although it might be theoretically appealing to unify DP phasehood and Specificity, or to eliminate one of them altogether, having only one constraint is not empirically adequate (contra Matushansky 2005; Huang 2022) when cross-linguistic patterns are considered.

Table 2: Predictions and experiment results

	DP phasehood +Incorporation	Specificity Condition	Results
Definite island effect in English	non-VOCs only	non-VOCs and VOCs	non-VOCs and VOCs (smaller with VOCs)
VOC effect in English	Yes	No (all else being equal)	Partial
Definite island effect in Chinese	No	non-VOCs and VOCs	non-VOCs and VOCs
VOC effect in Chinese	No	No (all else being equal)	No

We propose that both the DP phasehood approach and the Specificity Condition are needed to account for the pattern we observed. *Wh*-questions in English are constrained by the PIC as they involve *wh*-movement, and they are also constrained by the Specificity Condition since the moved *wh*-phrase binds its trace. On the other hand, *wh-in-situ* questions in Chinese are only restricted by the Specificity Condition since the *wh*-element is bound *in-situ* by a question operator. The PIC is not applicable, since no movement is involved. In this regard, we concur with standard analyses of Chinese *wh-in-situ*, such as Li 1992; Aoun and Li 1993; Tsai 1994.

For English subextraction from definite DPs, both the PIC and the Specificity Condition are violated when the main verb is a non-VOC: movement of the *wh*-element violates the PIC, while the binding of the *wh*-trace inside the specific DP violates the Specificity Condition (25a). The violation of both constraints results in a stronger perception of unacceptability. In contrast, when the main verb is a VOC, N/D-incorporation applies, effectively neutralizing the phasehood of the definite DP, thus ensuring that subextraction does not violate the PIC. On the other hand, incorporation has no impact on the Specificity Condition, since the *wh*-element still binds its *wh*-trace inside the definite DP. As a result, we still observe some degradation in acceptability, albeit to a smaller extent than with non-VOCs (25b).

- (25) a. What did Tom read [_{DP=phase} that book about t_{what}]? (*movement, *binding)
 b. What did Tom write [_{DP≠phase} that book about t_{what}]? (✓movement, *binding)

For Chinese, we propose that the definite island effects observed with VOCs and non-VOCs reflect a violation of the Specificity Condition, in which a question operator binds an *in-situ* *wh*-phrase inside a specific DP. Since movement is not involved in these *wh*-questions, there is no PIC violation. Given this proposal, N/D incorporation onto VOCs, even if it exists in Chinese, would not have made a difference, as incorporation only affects phasehood and movement but not on the Specificity Condition or binding (26). This explains why no VOC effect is observed in Chinese, i.e. why the size of the definite island effect remains the same across verb types.

- (26) Gloss for sentences in Chinese corresponding to (25)
 a. $OP_{[+Q]i}$ Tom read-ASP [_{DP} that about $what_i$ DE book]? (movement: n/a, binding: *)
 b. $OP_{[+Q]i}$ Tom write-ASP [_{DP} that about $what_i$ DE book]? (movement: n/a, binding: *)

Table 3 summarizes the constraints that are involved in each condition under discussion so far.

Table 3: Summary of violations

Condition	Example	PIC	Specificity Condition
non-VOC in English	... what Tom read that book about __	Violated	Violated
VOC in English	... what Tom wrote that book about __	Satisfied	Violated
non-VOC in Chinese	... Tom read that about what DE book	n/a	Violated
VOC in Chinese	... Tom wrote that about what DE book	n/a	Violated

We note that our proposal for English assumes constraint-stacking – the more violations there are in a sentence, the less acceptable the sentence – an idea that has been explored often in theories of locality (see Chomsky 1973:239 for an early discussion, also in the context of subextraction in English). Our contribution here is to present a concrete theory of constraint-stacking, by spelling out the constraints involved and supporting it with experimental evidence from both overt *wh*-movement and *wh-in-situ* languages.

We note that our results raise an interesting quantitative question for our proposal. Namely, in our analysis, subextraction with VOCs in English and *wh-in-situ* in Chinese DPs (VOCs and non-VOCs alike) both violate only one constraint – Specificity. Yet the violation incurs a larger penalty in Chinese than in English (~1 vs. ~0.2 z-units respectively).

This effect size difference, strictly speaking, does not contradict any part of our proposal. Our proposal is intended to account for the presence or absence of certain effects, but not the magnitude of these effects across languages. In fact, as far as we know, no syntactic theory makes clear predictions about whether the effect size of a given constraint should vary cross-linguistically. Cases of the same constraint inducing effects of different sizes across languages have been reported in other island constructions. For example, Almeida (2014) reports that *wh*-island effects are smaller in Brazilian Portuguese than in English, to the extent that he

calls the effect in Brazilian Portuguese ‘subliminal.’ He suggests that ‘the acceptability penalties that are incurred when these putative universal constraints are violated simply vary across languages’ (p. 69). Our experiment observes a similar case of cross-linguistic variation where violating the Specificity Condition induced a larger penalty on acceptability in Chinese than in English. Ultimately, this effect size difference may reflect some difference between the *wh*-dependencies of the two languages, since the English and Chinese items were deliberately constructed to be parallel in lexical semantics and pragmatics. For instance, perhaps acceptability is sensitive to the category of the binder: a *wh*-phrase in English, a question operator in Chinese.

4.2 PIC and binding

Our proposal assumes that the locality constraints on *wh-in-situ* questions and *wh*-fronting questions are distinct: while both overt movement and *wh-in-situ* are subject to the Specificity Condition, overt movement is subject to PIC and binding is not. One might question this assumption, however: according to standard definitions of the PIC, operations cannot cross a phase boundary. If so, why is binding possible when the operator and the variable are not in the same phase?

We argue that it is possible to make sense of this by adopting the bottom-up cyclic linearization theory of the PIC proposed by Fox and Pesetsky (2004). According to Fox and Pesetsky, when elements within a phase are Spelled-out to PF, they remain syntactically active but the linear order of these elements are fixed at the point of Spell-out. Importantly, this order cannot be changed in the course of a derivation. Fox and Pesetsky (2004) argue that because of this order preservation property, an element that overtly moves out of a phase must first stop at the phase edge, as required by the PIC.

To illustrate, the examples in (27) show two derivations of the English *wh*-question *What do you think Mary would eat?*.

- (27) What do you think Mary would eat?
- a. What_i do you think [_{CP} ~~what~~ Mary would eat ~~what~~]?
 - b. What do you think [_{CP} Mary would eat ~~what~~]?

In (27a), the object *what* first moves to the edge position of the embedded CP phase, consistent with the PIC. The CP then undergoes Spell-out, fixing the order of the elements in this phase. In particular, *what* precedes other elements in the embedded CP. *What* subsequently moves to the matrix CP. Crucially, the order established in the lower CP has not changed: *what* still precedes every element in the lower CP. The derivation can proceed without a problem.

In the derivation in (27b), *what* in the embedded CP remains *in-situ* when this CP undergoes Spell-out. *What* is then stipulated to follow all other elements within this phase. After subsequent operations create the matrix CP, *what* moves directly to Spec,CP, contrary to the PIC. The cyclic linearization theory explains why this derivation is problematic: when the matrix CP undergoes Spell-out, *what* ends up preceding all elements, including elements in the embedded CP. As a result, this derivation runs into an ordering conflict: *what* both follows and precedes the elements in the embedded phase.

In other words, this order preservation theory provides a way of understanding why overt movement must obey the PIC: crossing a phase boundary in a single movement operation (in

violation of the PIC), as in (27b), would result in an ordering conflict. Operations that do not change word order, such as string-vacuous movement (Fox and Pesetsky 2004 fn. 9), are free to cross phase boundaries, i.e. violate the PIC. The assumption that binding is not sensitive to the PIC is thus compatible with this approach to phasehood.

4.3 Experiment 3: Mandarin Chinese *wh*-indefinites

Our proposal crucially relies on prior proposals about binding and *wh*-phrases in Chinese. One such proposal is Li 1992, which reports that Chinese *wh*-phrases can have an indefinite reading, but not when the *wh*-phrase appears inside a demonstrative-marked nominal. Arguing that the indefinite reading arises when a *wh*-phrase is bound by an operator, Li attributes this sensitivity to demonstratives to the Specificity Condition: a demonstrative-marked nominal is specific (in the sense intended by Fiengo and Higginbotham, rather than, say, Cheng and Sybesma), and so blocks binding. In the review process, anonymous reviewers expressed concerns about the robustness of Li's observations, which call into question her proposal about binding and the Specificity Condition, in turn potentially undermining the logic behind our Experiments 1 and 2. To the best of our knowledge, no prior experimental syntax work has been done on these *wh*-indefinites. We therefore conducted a third experiment over the internet using PCIBex, to confirm the (un)acceptability of *wh*-indefinites within definite and indefinite DPs. If the experiment bears out Li's observations, this would support the binding-based analysis of *wh*-indefinites.

4.3.1 Design

Li reported a number of *wh*-indefinite examples showing this sensitivity to specificity. For practical reasons, we based our materials on only one of her examples, reproduced below as (28).

- (28) Wǒ yǐwéi tā ná-le (*nà-zhāng) shénme rén de xiàngpiàn.
 I mistakenly.believe he take.away-PERF that-CL what man de picture
 Intended: "I mistakenly thought he took away (a/that) picture of someone."
 (adapted from Li 1992:148 ex. 54)

Accordingly, our materials had four conditions, manipulating the definiteness, and therefore the specificity, of the object and the presence/absence of a *wh*-indefinite in that object in a 2x2 design. The two *wh*-indefinite conditions (29a) and (29b), differing minimally in the presence of the indefinite *yī-CL* 'one-CL' and the demonstrative *nà-CL* 'that-CL' in the object, are directly aimed at reproducing the contrast reported in (28), while reusing the same definiteness/specificity manipulation in Experiments 1 and 2. However, we also wanted to make sure any contrast observed would not just be a more general contrast between the indefinite *yī* 'one' and the demonstrative *nà* 'that.' For this reason, we had two more conditions (29c) and (29d), replacing the *wh*-indefinite with *mǒu-CL* 'a certain N' or 'some N'. *Mǒu* ensures that the inner DP remains indefinite, thus maximizing comparability with the *wh*-indefinite cases. At the same time, to the best of our knowledge, there is no claim that *mǒu* involves any kind of binding relation. We therefore expect it to be insensitive to the specificity or definiteness of the object.

- (29) a. Wǒ hái yǐwéi Qián Tíng ná-le yì-zhāng nǎ-wèi
 I part mistakenly.believe Qian Ting take.away-PFV one-CL which-CL
 lǎoshī de zhàopiàn.
 teacher de picture.
 ‘I mistakenly thought Qian Ting took away a picture of some teacher.’
 (nondef | *wh*-indef)
- b. Wǒ hái yǐwéi Qián Tíng ná-le nà-zhāng nǎ-wèi
 I part mistakenly.believe Qian Ting take.away-PFV that-CL which-CL
 lǎoshī de zhàopiàn.
 teacher de picture.
 ‘I mistakenly thought Qian Ting took away that picture of some teacher.’ (def
 | *wh*-indef)
- c. Wǒ hái yǐwéi Qián Tíng ná-le yì-zhāng mǒu-wèi
 I part mistakenly.believe Qian Ting take.away-PFV one-CL certain-CL
 lǎoshī de zhàopiàn.
 teacher de picture.
 ‘I mistakenly thought Qian Ting took away a picture of a certain teacher.’
 (nondef | *mǒu* ‘certain’)
- d. Wǒ hái yǐwéi Qián Tíng ná-le nà-zhāng mǒu-wèi
 I part mistakenly.believe Qian Ting take.away-PFV that-CL certain-CL
 lǎoshī de zhàopiàn.
 teacher de picture.
 ‘I mistakenly thought Qian Ting took away a picture of a certain teacher.’
 (def | *mǒu*)

To keep the experiment relatively short and avoid participant fatigue, we opted for a between-participant design, such that each group saw and rated sentences from two out of the four conditions: either the *wh*-indefinite conditions or the *mǒu* conditions. We opted for what Marty et al. (2020) called joint presentation, in which two target sentences were presented together on the same screen, one from each condition. The two sentences were minimal pairs, identical except that one had the indefinite (non-specific) *yī-CL* and the other the definite (specific) *nà-CL*; the ordering of the two sentences was randomized. Participants were asked to judge each sentence for acceptability. We note that joint presentation is not commonly used in experimental syntax, which typically presents sentences one at a time in a pseudo-randomized order. However, as documented by Marty et al. and also reflected in our own initial testing, the traditional mode of presenting sentences one at a time is less sensitive than joint presentation, typically producing smaller effect sizes and requiring much larger number of participants. We further note that while joint presentation is not commonly used in experimental syntax, it very much resembles the presentation and judgment of minimal pairs in informal introspection, which is commonly used in syntactic theory research.

4.3.2 Materials

We created 10 lexical frames based on (29), generating 10 pairs of sentences that contained a *wh*-indefinite and another 10 pairs that contained a *mǒu-CL N* ‘a certain N.’ Each pair differed

in whether the object was indefinite, marked with *yī* ‘one,’ or definite/specific, marked with the demonstrative *nà* ‘that.’

The 10 *wh*-indefinite pairs were assigned to one list and the other 10 *mǒu-CL N* pairs to a second list. Each list was then combined with a common set of 20 filler items. Each filler item consisted of a pair of sentences that are minimally different, with two exceptions: in these two filler items, one of the two sentences was an attention check and labeled as such. For these sentences, participants were told to drag the slider all the way to the left or to the right.

All else being equal, joint presentation might cause participants to believe that one sentence should be rated as distinctly more acceptable than the other. To avoid this misconception, we created three practice pairs of sentences, which were presented before participants started the experiment proper. In the first pair, one sentence was acceptable and the other was not. In the second pair, both sentences were both ungrammatical and unacceptable, and in the third, both sentences were grammatical and acceptable. We added explicit comments on the (un)acceptability of these practice items. To reinforce this message, we made sure that of the 20 filler items, five items were such that both sentences were ungrammatical and unacceptable. Another five items were such that both sentences were grammatical and acceptable.

4.3.3 Participants

40 participants were recruited over Prolific. All were self-reported native speakers of Mandarin Chinese. Each Prolific participant received GBP 1.50 upon completing the experiment.

4.3.4 Procedure

Participants accessed the experiment by visiting PCIBex. Participants first saw the three practice items; they were instructed to use a slider to indicate the acceptability of each sentence. Half the participants were assigned to the first list, where the target items were *wh*-indefinite pairs; the other half were assigned to the second list, where the target items were the *mǒu-CL N* pairs. Filler and target items were presented in a pseudo-randomized fashion, so that target items were never presented consecutively.

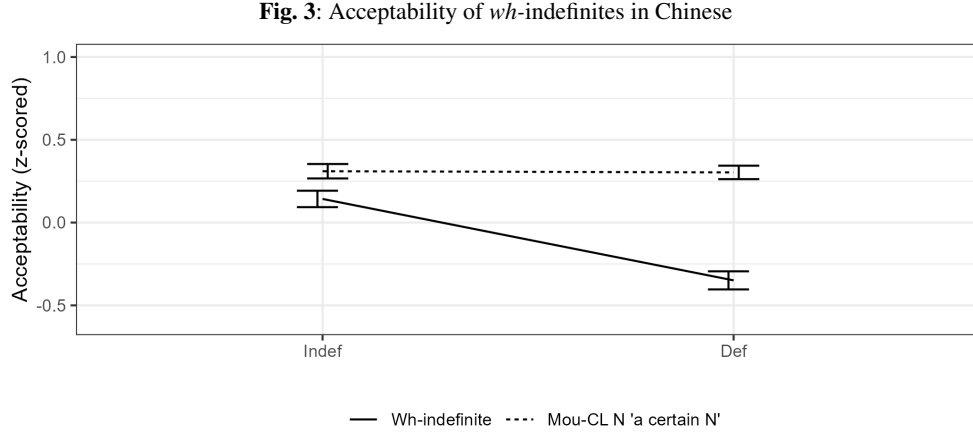
4.3.5 Data analysis

Participant ratings were again standardized using the z-score transformation. All participants passed the attention checks, so we did not exclude any participants from analysis.

We model z-scored acceptability ratings of target items using linear mixed effects models. Reflecting the 2×2 design of the experiment, the model had two predictors: the definiteness (or specificity) of the object and the presence/absence of a *wh*-indefinite in that object (where indefinite objects were coded as 0, definite objects as 1; the presence of *mǒu* and not a *wh*-indefinite coded as 0, and the presence of *wh*-indefinite as 1). We started with the maximal model and reduced the random effects structure until the model converged; the final model included by-participant random intercepts and random slopes for both predictors.

4.3.6 Results

Figure 3 presents z-scored acceptability ratings for the four conditions. All conditions had positive ratings, except for the condition with definite (demonstrative-marked) objects containing *wh*-indefinites. These were much less acceptable, with negative ratings. Statistically speaking, there was a significant interaction effect between the presence/absence of a *wh*-indefinite and the definiteness (or specificity) of the object containing it ($\beta = -0.48$, *s.e.* = 0.15, $t = -3.32$, $p < .01$).

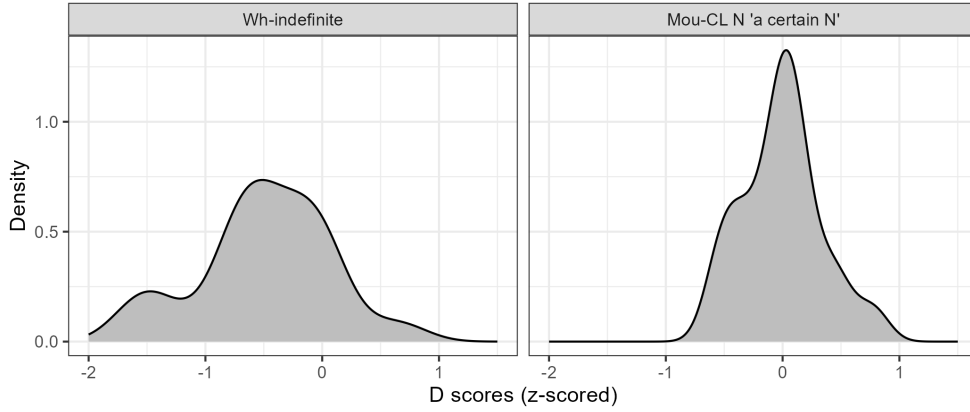


Note: For reference, in terms of mean z-scored acceptability ratings, our filler items had a range of -1.44 to 0.92 ; unacceptable fillers had a mean rating of -0.48 and acceptable fillers had a mean of 0.37 .

Again, for thoroughness, we considered the possibility of variability within participants. Because of our between-participant design, we could not calculate a DD score for each participant; this would require each participant to have seen and rated all four conditions. Instead, we calculated a D(ifference) score, taking a participant's mean ratings for the definite (demonstrative-marked) items and subtracting their mean ratings for the indefinite items. The distribution of D scores are shown in Figure 4. For the *wh*-indefinite items, which by hypothesis involve binding, D scores generally are distinctively negative, showing that participants on the whole found definite items to be less acceptable than indefinite ones.

In other words, we found the contrast reported by Li for *wh*-indefinites; in fact, our two *mǒu-CL N* conditions further confirmed that the contrast is restricted to *wh*-indefinites. This finding therefore validates Li's (1992) claim about Chinese *wh*-indefinites being sensitive to specificity. This in turn provides novel experimental support for her analysis – which we adopt – that the interpretation of *wh*-phrases is dependent on binding by an operator, which is subject to the Specificity Condition.

Fig. 4: Distribution of D scores in Chinese



Note: D score = rating for definite (demonstrative-marked) items – ratings for indefinite items.

5 General discussion

To recap preceding sections, the DP phasehood and Specificity Condition make different predictions about the acceptability of subextraction of *wh*-arguments from definite DPs and *in-situ wh*-arguments inside definite DPs. We evaluated these predictions using Experiments 1 and 2. We found a definite island effect for both VOCs and non-VOCs in English. Additionally, the effect is smaller in sentences with VOCs. In other words, the presence of VOCs made subextraction out of the demonstrative-marked definite DPs more acceptable, although it did not completely neutralize the definite island effect. For Mandarin Chinese, the definite island effect is comparable in size for both VOCs and non-VOCs, suggesting that there is no VOC effect in Chinese. We proposed that these results can be explained if both DP phasehood/the PIC and the Specificity Condition are part of a general theory of locality.

A critical ingredient in our account is the idea that binding is generally sensitive to the Specificity Condition, following Li 1992. We therefore ran Experiment 3 on Chinese *wh*-indefinites, which Li claims also involve binding. Results from this experiment showed that *wh*-indefinites are more acceptable inside indefinite objects compared with definite ones. These results are consistent with Li’s observations, parallel our Chinese *wh-in-situ* results from Experiment 2, and more importantly, provide support for our adoption of her proposal.

These results bear on a number of theoretical issues. To preview the rest of this section, we suggest that they pose challenges for information structure approach to locality. We also argue that these results provide support for a theory where *in-situ wh*-arguments are insensitive to a locality constraint like the PIC, contra Lu et al. (2020); we also discuss the relevance of the Specificity Condition in recent experiments on the island sensitivity of Chinese *wh-in-situ*.

5.1 On the information structure approach to locality

In the introduction, we briefly mentioned a third account of the definite island effect, namely, an information structure-based approach. This approach argues that (certain) extraction out

of definite objects results in a discourse clash thus making the sentence infelicitous. An influential early proposal is what Erteschik-Shir (1981:161) calls the Dominance Condition (DC), which states that “[a]n NP can only be extracted ... out of phrases in which the NP may itself be regarded as dominant” (also Erteschik-Shir and Lappin 1979). Briefly, a constituent is dominant if the speaker intends to direct hearers’ attention to it (Erteschik-Shir 1981:160). In essence, Erteschik-Shir argues, on the basis of various diagnostics, that the *wh*-phrase complement of a definite object under non-VOCs is not dominant. As a result, the *wh*-phrase cannot undergo subextraction, thus capturing the definite island effect, as shown in (30).

- (30) a. you listen to [_{notDominant} a song about who_{dominant}] (DC satisfied)
 b. *you listen to [_{dominant} that song about who_{notDominant}] (DC violated)

More recently, Abeillé et al. (2020) propose a Focus-Background Conflict (FBC) constraint (31) which predicts that “only if a construction puts the extracted element into focus, and if this element belongs to a backgrounded constituent, a discourse clash will occur” (p. 3). Their experiments showed that in English and French, extraction out of subjects, which are assumed to be backgrounded, is degraded in *wh*-questions but not in relative clauses.

- (31) Focus-background conflict (FBC) constraint: A focused element should not be part of a backgrounded constituent. (Abeillé et al. 2020:3)

Abeillé et al. (2020) claim that the definiteness effect can be accounted for by the FBC constraint since definite objects are backgrounded, and cite Grosu (1981) who claims that relative clauses do not show definiteness effects while *wh*-questions do in (32) (contra Ross 1967; Simonenko 2016; Shen and Lim 2022, also see discussion in the next section).

- (32) a. Which actress did you buy [a picture of]?
 b. #Which actress did you buy [that picture of]?
 c. That is the actress who I bought [a/that picture of].
 (Abeillé et al. 2020:20 ex. 39)

Another proposal that more closely resembles Erteschik-Shir 1981 is Goldberg 2006, which proposes that backgrounded constructions are islands (BCI), as in (33). Cuneo and Goldberg (2023) provide experimental evidence for the BCI, showing that the acceptability of extraction in *wh*-questions and relative clauses correlates with the backgroundedness of the constructions where the elements are extracted from in complex NP and adjunct islands, among others. While the BCI approach does not make direct reference to the definiteness effects discussed in this article, one can expect an analysis along lines similar to the Dominance Condition.

- (33) Backgrounded constructions are islands: Constructions are islands to long-distance dependency constructions to the extent that their content is backgrounded within the domain of the long-distance dependency construction. (Cuneo and Goldberg 2023:2)

We did not discuss these information structure proposals in the above sections, because we are not sure of the predictions of this approach for either English or Chinese. As far as we can tell, unlike Davies and Dubinsky 2003, Erteschik-Shir and other advocates of this approach do not explicitly discuss the VOC effect with demonstrative-marked definites (which we tested in our experiments).

In addition, a more basic problem for us is that existing proposals of this approach are mostly silent about *wh-in-situ* (except for brief remarks in Goldberg 2006). That said, we believe that our results pose problems for this approach, whether it covers *wh-in-situ* or overt *wh*-movement.

First, for the sake of argument, suppose that the information structure approach only applies to overt movement and not to *wh-in-situ*. In that case, the information structure approach does not predict the definite island effect observed in Chinese, and something like the Specificity Condition would have to be assumed. However, this might run contrary to the spirit of certain information structure accounts. In at least some proposals, information structure is implied as obviating the need for other constraints like subjacency (or PIC) or the Specificity Condition (e.g., Erteschik-Shir and Lappin 1979; Goldberg 2006; Ambridge and Goldberg 2008).

Alternatively, suppose that this approach covers both overt *wh*-movement and *wh-in-situ*, following Goldberg (2006:153-154). This is not an implausible position to take. A general idea behind this approach is that constraints on *wh*-dependencies are ultimately due to pragmatics and discourse function, and it is reasonable to assume that *wh*-questions have the same pragmatic properties and discourse functions cross-linguistically, even though their syntax might vary substantially. However, if this were the case, then we should predict uniformity in English and Chinese: to the extent that English has a VOC effect, so should Chinese. The absence of a VOC effect in our Chinese results, then, is puzzling for this approach, and again suggests that invoking some other principle is necessary.

Another limitation of these proposals, as far as we can tell, is that they are silent about *wh*-indefinites (cf. our Experiment 3). They therefore do not offer predictions about whether *wh*-indefinites are sensitive to definiteness (more precisely, specificity), in contrast to Li's (1992) proposal, which makes use of the Specificity Condition.

In sum, we did not set out to evaluate the information structure-based approach because we were uncertain about its predictions for English and Chinese. Given our experimental findings, though, we believe that our results present an empirical challenge for a theory of locality that is founded on information structure principles. This empirical challenge could be resolved if the theory is supplemented with additional constraints, much like how our proposal combined DP phasehood/PIC and the Specificity Condition. However, as we noted above, doing so might run counter to the spirit of existing information structure proposals.

5.2 On an LF movement analysis for *wh-in-situ*

In this section, we address the LF movement approach to *wh-in-situ*, an alternative to the unselective binding analysis that we adopt as part of our proposal. Two variants of the LF movement approach have been proposed. In its original formulation, C.-T. J. Huang (1982b) argues that *wh-in-situ* questions involve *wh*-movement in LF and LF movement of *wh*-arguments is not subject to subjacency.

If LF *wh*-movement of *wh*-arguments is not subject to subjacency or its Minimalist counterpart, the PIC, then it is unsurprising that there is no verb of creation effects in Chinese: it does not matter whether D incorporates or not onto a VOC. The definite island effects we observed for *in-situ wh*-arguments in our Chinese experiment must have some other source, such as the Specificity Condition. In other words, this particular LF movement analysis of

in-situ wh-arguments makes the same predictions as the unselective binding analysis. However, the unselective binding analysis covers more empirical ground; unlike the movement analysis, it can also account for the definiteness effect on *wh*-indefinites.

On the other hand, Lu et al. (2020) report an experiment demonstrating that complex NP island effects can be elicited with both *in-situ wh*-arguments and *wh*-adjuncts in Chinese. On this basis, they argue that LF movement is sensitive to a locality constraint like subjacency or the PIC, contra C.-T. J. Huang 1982b and many others.

At first glance, this particular version of LF movement analysis of *wh-in-situ* seems compatible with our experiment results. It is tempting to treat the definite island effect in both English and Chinese as a PIC/subjacency violation: we would no longer have to appeal to the Specificity Condition to account for the Chinese facts. The difference in the VOC effect for English and Chinese might be accounted for if one assumes parametric variation in N/D-incorporation, such that it is available in English but not in Chinese.

However, there are several reasons not to pursue this approach either. First, it assumes that there is no island effect in English VOC sentences. Our results show otherwise: there is a smaller definite island effect there. We would therefore still need the Specificity Condition to explain this effect in English VOC sentences as well as the Chinese *wh*-indefinite data that we verified experimentally. Since this particular LF movement analysis actually requires both PIC/subjacency and the Specificity Condition, it does not offer a clear theoretical or empirical advantage over our proposal in Section 4.1.

Second, like Tian et al. (2022), we believe that Lu et al.’s experimental results are confounded, which weakens the case for this analysis of LF movement. Specifically, Lu et al. provide acceptability judgment evidence showing that sentences like (34), with *in-situ wh*-arguments inside relative clause islands, are relatively unacceptable, not unlike equivalent sentences with *in-situ wh*-adjuncts, which are usually taken to be ungrammatical.

- (34) Yuèhàn xiǎngzhīdào Biěr jiàn-le chī shénme de nǚhái.
 John wonder Bill meet-PFV eat what DE girl
 ‘John wonders what Bill met the girl that ate.’ (adapted from Lu et al. 2020:615 Table 1)

Tian et al. (2022) argue that Lu et al.’s materials are pragmatically odd (see also Jin and Yan 2024 and Yue and Xiang 2025 for similar discussions): relative clauses tend to denote a prominent feature of the modified noun but “eating a certain thing is not the prominent feature of a girl” (ibid.:4). Tian et al. argue that it is this pragmatic oddness that drives the relative unacceptability of (34). In their own experiment, Tian et al. use stimuli that are said to be pragmatically more natural, like (35), where the type of prey is a prominent way of identifying hunters. Their results show that these sentences are more acceptable than equivalent sentences with *in-situ wh*-adjuncts; no superadditive effects were found for *in-situ wh*-arguments inside complex NPs. More recent experiments by Jin and Yan (2024) have also failed to find super-additive effects for *in-situ wh*-arguments inside a complex NP, contrary to Lu et al.’s (2020) proposal. Yue and Xiang (2025) found the super-additivity for *in-situ wh*-arguments to be marginal once contexts are provided, while the island effect for *in-situ wh*-argument remains significant.

- (35) Zhōuyǒng xiǎngzhīdào zhèngfǔ huì chéngfǎ bùshā shénme de lièrén.
 Zhouyong wonder government will punish hunt what DE hunters
 ‘Zhouyong wonders what is the thing x such that the government will punish the
 hunter who kills x .’ (Tian et al. 2022:8 ex. 27, rated around 6 on a 7-point scale)

We agree with Tian et al. (2022), Jin and Yan (2024), and Yue and Xiang (2025) that there are confounds in Lu et al.’s materials and that pragmatic naturalness plays a role. However, we also want to highlight the role of definiteness in this debate. Lu et al.’s materials contain *in-situ wh*-phrases inside “bare” DPs that are objects of verbs like *jiàn-le* “meet-PFV” in (34). According to Cheng and Sybesma (1999:525) (citing Sybesma 1992:176-178), bare nominal objects of bound predicates, such as *jiàn-le*, are interpreted as definite in Chinese. The definiteness of the object is also reflected in Lu et al.’s own translation of the DP object *chī shénme de nǚhái*: ‘the girl who ate [what]’ (our emphasis). If so, sentences like (34) violate the Specificity Condition. In other words, we think that what Lu et al. detected was (at least partially) a definite island effect.

On the other hand, based on the example stimuli in Tian et al. 2022 like (35), we believe that Tian et al.’s materials contain complex NPs that are likely to be interpreted as non-specific indefinites. Based on the discussion of Cheng and Sybesma 1999 and Sybesma 1992, bare nominal objects of unbounded predicates, such as *huì chéngfǎ* “will punish” in (35), can receive a non-specific, indefinite reading. The sentence in (35) therefore can also be plausibly translated as “Zhouyong wonders what is the thing x such that the government will punish hunters who kill x ”, i.e. where the government’s policy is against a general class of hunters (reflected in the generic “hunters” in our translation). If this non-specific, indefinite interpretation is also the one that their participants assigned to these stimulus sentences, there would be no violation of the Specificity Condition. This would explain the relative acceptability of (35).

In sum, we believe that the differences between Lu et al.’s and Tian et al.’s findings can be at least partially attributed to the Specificity Condition. If this interpretation is correct, these results would provide further experimental support for existing claims that *in-situ wh*-arguments are not sensitive to subjacency/PIC but are sensitive to the Specificity Condition, in line with our findings here.

5.3 Remarks on *wh*-adjuncts

We note that we have restricted our discussion in this paper to movement of *wh*-arguments and *in-situ wh*-arguments. However, as a reviewer pointed out, our proposal is couched in terms of *wh*-dependencies more generally, and therefore has implications for *wh*-adjuncts as well.

This is true not only for our proposal but also the works we cited. In our case, our decision to focus on *wh*-arguments is based on both theoretical and practical considerations. First, our experiments directly build on observations made in the literature on subextraction from DPs. These observations have consistently featured *wh*-arguments but not *wh*-adjuncts; this includes both Davies and Dubinsky 2003 (for an early version of the PIC/incorporation account) and Fiengo and Higginbotham 1981 (for the Specificity Condition), as well as Simonenko 2016; Erteschik-Shir 1981. For this reason, we have done the same here.

Second, while our proposal has implications for *wh*-adjuncts, there is uncertainty over exactly what predictions our account (or the PIC or Specificity Condition) makes for *wh*-adjuncts, because *wh*-arguments and *wh*-adjuncts are not subject to exactly the same constraints. This is the case both for *wh*-movement languages as well as *wh-in-situ* languages. In English, extracting adjuncts out of DPs is observed to be unacceptable, regardless of definiteness as shown in (36) (Huang 1982b:564, also Chomsky 1986a:80ff.). The different locality restrictions on *wh*-arguments and *wh*-adjuncts in *wh*-movement languages partly motivated the Empty Category Principle in the Government and Binding framework. If an experiment on definite DP islands were conducted with *wh*-adjuncts, the effects of definiteness and verbs of creation are likely to be affected by the general ban on extracting adjuncts out of DPs, rendering the results to be hard to interpret.

- (36) a. Of whom did you buy the picture? (Huang 1982a:564, ex. 100)
 b. *From which city did you meet the men? (ibid., ex. 102)

In *wh-in-situ* languages like Chinese, it has been observed that *wh*-adjuncts denoting manner (e.g., *zěnmé* ‘how’) and reason (e.g., *wèishénme* ‘why’) show sensitivity to syntactic islands, unlike *wh*-arguments (see Huang et al. 2009 for a review). This asymmetry has been experimentally verified for complex NP, subject, and adjunct islands (Tian et al. 2022; Jin and Yan 2024; Yue and Xiang 2025). The sentences in (37) illustrate that *in-situ wh*-adjunct *wèishénme* ‘why’ is unacceptable inside a complex DP while *in-situ wh*-argument *shénme* ‘what’ is acceptable.¹⁴

- (37) a. Zhōuyǒng xiǎngzhīdào zhèngfǔ huì chéngfǎ bùshā shénme de lièrén.
 Zhouyong wonder government will punish kill what DE hunter
 ‘Zhouyong wonders what is the thing *x* such that the government will punish the hunter who kills *x*.’ (Tian et al. 2022, ex. 27)
 b. *Zhōuyǒng xiǎngzhīdào zhèngfǔ huì chéngfǎ wèishénme bùshā jīngyú de lièrén.
 Zhouyong wonder government will punish why kill whales DE hunter
 ‘Zhouyong wonders what is the reason *x* such that the government will punish the hunter [who kills whales for *x*].’ (Tian et al. 2022, ex. 31)

The sensitivity of *in-situ wh*-adjuncts to islands has motivated an analysis where they undergo covert movement to Spec,CP in Mandarin Chinese (e.g. Tsai 1994). As suggested by a reviewer, if this analysis is on the right track, one would expect that *wh*-adjuncts show the definite DP island effect (assuming that covert movement is subject to the PIC) and perhaps even a VOC effect (if Chinese has the same conditions on N/D incorporation as English). However, an experiment testing the definite DP island effects with these *in-situ wh*-adjuncts would require sentences with a clausal modifier to the noun that can host adjuncts denoting manner or reason. A DP with such a clausal modifier is by definition a complex DP, which, as shown in (37), is a syntactic island independent of its definiteness. The results of an experiment probing the definite DP island with *wh*-adjunct in Chinese would be hard to interpret as the Complex NP Constraint would also be violated.

¹⁴In the translations of (37), taken from Tian et al. 2022, the complex DP is definite and specific, but the examples could also have an indefinite, non-specific interpretation, as we noted as in the previous section.

Due to these issues, we (and we suspect existing proposals) limit the discussion to *wh*-arguments. Of course, this is not to claim that it is impossible to test *wh*-adjuncts and DPs, but currently we do not see a clear way and so we will leave this aside for future research.

6 Conclusion

This paper used experimental methods on English and Mandarin Chinese to explore theories of constraints on *wh*-dependencies involving definite nominals. Focusing on demonstrative-marked definite nominals, we observed a definite island effect in both *wh*-movement questions in English and *wh-in-situ* questions in Chinese. We also detected a VOC effect only in English, although the effect was less strong than what Davies and Dubinsky (2003) and Huang (2022) suggested. We proposed that a combination of two constraints – the PIC and the Specificity Condition – can account for the patterns observed, where the PIC restricts *wh*-movement and the Specificity Condition restricts binding in both *wh*-movement and *wh-in-situ* questions. The VOC effect results from head incorporation neutralizing the phasehood of DP, which ultimately affects movement, but not binding. We also presented experimental results showing that Chinese *wh*-indefinites are sensitive to definiteness, which provides independent support for our adoption of the Specificity Condition (see Li 1992). In addition, our results present challenges to some information structure based theories to locality and a LF movement analysis for *wh-in-situ* questions. Finally, our proposal also has implications for the syntax of nominal phrases. If our proposal is on the right track, the definite island effect in English can be due to a PIC violation, which in turn provides support for the view that DPs are phases.

Methodologically, our study provides further evidence of how experimental work can inform and enrich our theories. Our proposal – that a general theory of locality needs both the PIC/DP phasehood and the Specificity Condition – rests on our finding that the choice of main verb affects the size of the definite island effect in English but not in Chinese. This empirical finding was made possible by the factorial design of our experiments, which allowed us to detect fine-grained differences in acceptability more reliably.

While our findings have led us to a clearer picture of how definiteness and *wh*-dependencies interact in overt *wh*-movement and *wh-in-situ* languages, we wish to point out that our conclusion is by no means the full picture. For both scope and practical reasons (see section 3.2), we have only focused on demonstrative-marked DPs in our study. But as noted in the introduction, there are other subclasses of definite DPs, and it has been reported that some of them exceptionally do not show a definite island effect in English and other languages (Erteschik-Shir 1981; Fiengo 1987; Davies and Dubinsky 2003; Simonenko 2016; Huang 2022; Lim 2022, etc.). Future work could look to experimentally validate these reports, using the same factorial design we used for demonstratives, and to explore the extent to which our two-constraint framework accounts for the facts for these other subclasses of DPs.

Our findings also have consequences for island constraints more generally. Although widely acknowledged as a factor affecting subextraction from content noun nominals, definiteness and specificity are not always well-controlled for in work exploring other classic island constraints. In Section 5.2, we discussed two recent studies involving Chinese complex NPs (Lu et al. 2020; Tian et al. 2022), where it can be tricky to control for specific or definite readings, since Chinese lacks definite articles equivalent to English *the*. But similar comments apply to studies on English as well. In particular, work that target complex NP and subject

islands often use examples where the nominal phrases of interest are definite. (38a) provides an example from an experiment investigating English complex NP islands (among other islands), while (38b) give an example from an experiment probing for the subject island effect. Even supposing that these sentences do contain violations of complex NP and subject islands, our results raise the possibility that the observed effect might at least be partially driven by the definite island effect, in which case we would have overestimated the actual effect associated with complex NP and subject islands, with potential consequences for theories of these islands. In light of our results, future work on these constraints should systematically control for specificity and/or definiteness, so as to obtain a more accurate empirical picture.¹⁵

- (38) a. What did you hear [the statement that Jeff baked ___]? (Sprouse et al. 2016:318 ex. 14d)
 b. Who do you think [the gift from ___] prompted the rumor about the Senator? (ibid. ex. 15d)

Finally, returning to the present study, we would like to address the issue of parsimony – or perhaps more precisely, the apparent lack thereof – in our proposal that the definite island effect reflects (at least) two distinct constraints: the PIC and the Specificity Condition. While this proposal is perhaps not as parsimonious compared to what it could have been, these two constraints arguably reflect two different aspects of definiteness: the PIC arises because of how definiteness is structurally encoded in the form of DPs, while the Specificity Condition is clearly related to the semantics of definiteness. This view is also very much compatible with long-standing generative views on locality, which have recognized that acceptability judgments are a composite of structural and non-structural factors (Chomsky 1965; Schütze 1996; Sprouse 2007; Huang et al. 2025, and many others). Seen from this perspective, the two constraints are independent of each other. If so, even though simplicity is a virtue more generally speaking, there is little reason to expect that we can unify the constraints or eliminate one of them.

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¹⁵Of course, doing so might not be easy. As an anonymous reviewer quite rightly pointed out, it might be challenging to construct indefinite counterparts to the examples in (38). But if one of our goals is indeed to, say, tease apart the complex NP island effect from the definite island effect, then we should seriously consider using materials that are compatible with indefinite determiners.

Ethics approval of the experiments reported in the paper was granted by Departmental Ethics Review Committee (DERC) at the Department of English, Linguistics and Theatre Studies of National University of Singapore. Informed consent was obtained from all individual participants.

Materials of all four experiments are available at https://osf.io/9au5x/overview?view_only=656d01f7f4a7436b9276f55615a35411.

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Appendix A A follow-up experiment in Mandarin Chinese

An anonymous reviewer suggested that our Chinese *wh-in-situ* experiment (Section 3) might have a structural ambiguity that affected the interpretation of our results, where *in-situ wh*-arguments inside definite DPs always showed an island effect. More specifically, in that experiment, the *wh*-arguments were always inside a PP (in square brackets), which appeared together with a *de* marker, followed by the head noun, as illustrated in (39). The reviewer was concerned that this PP-*de*-Noun structure might instead have been (unintentionally) parsed as a complex DP with a relative clause, which also modifies its head noun with a *de* marker.

- (39) Dàijiā xiǎngzhīdào Shěn-lǎoshī jiǎng-lè nǎ-gè [guānyú shénmè] de xiàohuà.
 everyone wonder Shen-teacher say-PFV that-CL about what DE joke
 ‘Everyone wonders what Teacher Shen told that joke about.’

According to the reviewer, it is possible that the observed island effect came about only because participants had to process a (long) covert *wh*-dependency together with relative clauses and definiteness: processing all three simultaneously in a single sentence imposed demands that were large enough to overload the parser, resulting in what looked like an island effect. If the unintended relative clause parse could have been removed from the picture – although we cannot do so in reality, because PPs can only combine with their head nouns in Chinese in the presence of *de* – we might have found that the parser could easily process the *wh*-dependency and definiteness. If that is the case, we should have concluded that there is no definite island effect, and that the Specificity Condition does not hold in Chinese.

Following the reviewer’s suggestion, we tested this alternative hypothesis empirically by combining the same PP-*de*-noun sequence with another syntactic feature commonly associated with island effects, namely, subjects. More specifically, we ran a follow-up experiment using a 2×2 factorial design including four conditions shown in (40). The first factor was DP complexity: whether the DP that includes the *wh*-element is simple (*shénme N* ‘what/which N’) or complex, with a PP-*de*-noun structure (*guānyú shénme de N*, lit., ‘about what *de* N’). The second factor was DP function: whether this DP was in an object position or whether it was passivized to become the subject. We collected acceptability ratings and calculated a DD score, defined as ((40c)-(40d))-((40a)-(40b)).

- (40) Conditions for the follow-up experiment
- a. Wǒ xiǎng-zhīdào dúzhě-men zuì xǐhuān shénme shū.
 I want-know reader-PL most like what book
 ‘I wonder what book do readers like the most.’ (simple DP | object)
 - b. Wǒ xiǎng-zhīdào dúzhě-men zuì xǐhuān guānyú shénme de shū.
 I want-know reader-PL most like about what DE book
 ‘I wonder what do readers like books about the most’ (complex DP | object)
 - c. Wǒ xiǎng-zhīdào shénme shū zuì shòu dúzhě-men xǐhuān.
 I want-know what book most by reader-PL like
 ‘I wonder what books are liked by the readers the most.’ (simple DP | subject)
 - d. Wǒ xiǎng-zhīdào guānyú shénme de shū zuì shòu dúzhě-men xǐhuān.
 I want-know about what DE book most by reader-PL like
 Intended: ‘I wonder books about what are liked by the readers the most.’
 (complex DP | subject)

The logic behind this experiment is as follows. Instead of combining a complex DP (PP-*de*-Noun) with definiteness, we now combine it with being a subject while controlling for definiteness: the most natural interpretation of these complex DPs is a generic/indefinite reading. From an island perspective, subjects are similar to definiteness in two ways. First, both subjects and definite DPs are commonly said to be islands in many languages. Second, even though subjects are closely associated with island effects, they are not in Chinese: *in-situ wh*-arguments are acceptable inside subjects in Chinese (Huang 1982b, see also Jin and Yan 2024 for experimental evidence). This has a parallel with how *in-situ wh*-arguments themselves might be acceptable inside Chinese definite DPs, as posited by the alternative hypothesis.

If there is indeed a general parser overload effect, as predicted by the alternative hypothesis, we expect to see an interaction between DP complexity and DP function, and the DD score should be larger than zero. If, on the other hand, there is no such overload effect – consistent with a scenario where Chinese has a definite island effect – then no such interaction is expected and the DD score should be close to zero.

Materials

16 lexically matched sets were constructed for each condition and the test items were distributed using a Latin Square design. In particular, there were 16 distinct verb–noun pairs, consisting of 8 verbs and 16 nouns. The nouns in the materials were manipulated such that they should receive generic/indefinite readings.

Each participant saw 4 items per condition along with 32 filler items, as well as 2 attention check items and 2 grammatical check items. Assuming that all test items would receive relatively high ratings (i.e. no interaction effect), we had 24 fillers that were relatively unacceptable and 8 that were relatively acceptable. This was so that participants would encounter an equal number of acceptable and unacceptable sentences, as was the case in the original experiment. The procedure was identical to the Chinese experiment reported in Section 3.4: participants were asked to rate the naturalness of the sentences on a 100 point slider.

Participants

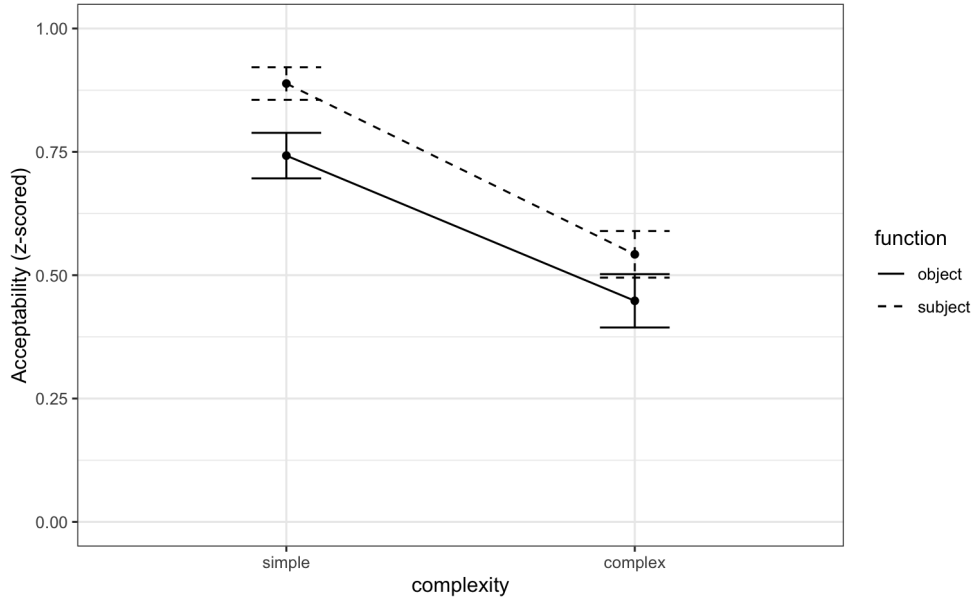
40 self-reported native speakers of Mandarin Chinese recruited from Prolific finished the experiment and responses from 1 participant were excluded from the analysis due to incorrect responses to one attention check item. Our analysis therefore included responses from 39 participants. On average, the experiment took 6 mins and each participant was paid 2 GBP.

Results and discussion

The mean zscore-transformed rating for each condition is shown in Figure A1. Overall, complex DPs are less acceptable, as is expected. But the critical complex DP subject condition still received relatively high ratings, comparable to the complex DP object condition. In z-score terms, the DD score is 0.05, very close to zero.

A linear mixed effect model was constructed with complexity and function as predictors as well as by-participant random slopes for complexity and function, and by-item random intercepts and random slopes for function (following the same procedure of starting with the maximal model and reducing the random effects until convergence). We found a significant

Fig. A1: Acceptability of subject island in Chinese



main effect of complexity and a significant main effect of function. Crucially, no significant interaction was found ($\beta = -0.07$, $s.e. = 0.08$, $t = -0.82$, $p = .41$).

The results do not support the proposed confound. Since there is no interaction between the DP complexity and the DP being a subject, the interaction we found in our original Chinese experiment is unlikely to be due to the parser overload predicted by the alternative hypothesis. This can be seen as providing (indirect) support for our original argument, in which Mandarin Chinese has a definite island effect.

We note that in this experiment *wh-in-situ* in the object is less acceptable than *wh-in-situ* in the subject as shown by the main effect of function. We speculate that this effect results from the dependency length difference between the two levels. When the parser encounters the question-embedding predicate *xiǎng-zhīdào* ‘want to know’, it initiates a search for a question word in the embedded clause (be it a *wh*-element or a word/phrase indicating a yes/no question). In the subject conditions (40c)-(40d), the parser encounters a *wh*-element shortly after: *shénmè* ‘what’ appears early in the embedded clause. In the object conditions ((40a)-(40b)), the search is resolved only later, when the parser encounters the *wh*-argument inside the object. We speculate that the longer search process in the object condition adds to the processing load, driving down the acceptability of the two object conditions (see Xiang et al. 2015 for the processing mechanisms of *wh-in-situ* questions in Mandarin Chinese).